The 3/2 Phase Transition in Semi-Random 2-SAT

0 Plan

1. Section 1: intro, plan, notations
   - introduce general framework
   - random 2 SAT
   - high level overview of our proof
2. Section 2: basic results
   - warm up: number of expected solutions
   - Minimal graph with positive UNSAT probability
   - Construction of the graph with density $m = 1.5 \cdot n$ edges and $\Pr[SAT] = 1$.
   - State and proof the theorem that if $m > 1.5 \cdot n$, then $\Pr[UNSAT] > 0$.
   - Observation that if graph G has many (k) small (constant size at most C) barbells, then the probability $\Pr[SAT] = e^{\Omega(-k/C)}$.
   - Discussion that graphs with many edge-disjoint barbells are easy. Then the next natural question is how to deal with hard graphs. What does it mean that a graph does not have many small barbells? Mention that there are graphs 3 or 4 regular with large girth $g(G) = \Omega(\log(n))$ that do not have even one small barbell. Such graphs are hard.
   - New subsection: Regularization of hard graphs. The purpose here is to get some useful regularity properties in case of hard graphs. Specifically, we apply a series of reduction arguments that only increase $\Pr[SAT]$ and keep positive excess and sufficiently large graph size.
     - More convenient description of easy graphs (those with many small edge-disjoint barbells): the graph has many small edge-disjoint cycles. State the graph-theoretic LEMMA that graph connected graph with k edge disjoint cycles (of any length) has at least $\lceil (k-1)/3 \rceil$ edge-disjoint barbells.
     - COROLLARY for the $\Pr[SAT] = e^{-\Omega(n)}$, when there are linearly many small cycles in total and all connected components have positive excess.
     - Reduce largest degree $\Delta(G)$ to a constant $\Delta_0 \geq \Delta(G)$ by splitting large degree vertices. This reduction has the property that $\Pr[SAT]$ and the number of vertices n may only increase, but excess would decrease, however we ensure that it is still positive and linear in n by choosing big enough $\Delta_0(\varepsilon)$.
     - Elimination of vertices of degrees 0 and 1. Obviously, it only may increase the excess and $\Pr[SAT]$ (probability does not change). It can only reduce the number of vertices by a constant factor. Make it an observation without any special environment (like lemma or claim)
     - Elimination of connected components with non positive excess. Again the excess and $\Pr[SAT]$ may only increase. Need only to check that we delete at most a constant fraction of vertices. Make it an observation.
     - Make one formal CLAIM summarizing the reduction of a hard graph so far: Each vertex has degree $2 \leq deg(v) \leq \Delta_0$; every connected component has positive excess; the overall excess is positive and linear in n.
     - Make another formal CLAIM stating: if G has at least $\rho_1 \cdot n$ vertices whose neighborhoods are not trees or at least $\rho_2 \cdot n$ loops, then G has ρ edge-disjoint short cycles. The proof is easy and short: in a constant degree graph short cycles at each of the vertices with non-tree neighborhood may only intersect (by edges) at most a constant $C(\Delta_0)$ of other such cycles.
     - Conclusion. We may assume that the "hard" graph has no loops, has degree of each vertex $2 \leq deg(v) \leq \Delta_0$, is locally tree-like with at most $\rho \cdot n$ vertices.
3. Section 3: Expected number of components in the solution graph

- Proof of the main identity

4. Section 4: the analysis for difficult graphs.

0.1 Notations

- Graph Theory notations (standard and basic)
  1. $G = (V, E)$ graph with multi-edges and loops, has $n = |V(G)|$ and $m = |E(G)|$.
  2. For the graph G and any other graph H, $C(H)$ – number of connected components.
  3. We also define cycle rank $\nu(G) \stackrel{\text{def}}{=\!=} |E(G)|-|V(G)|+c(G)$ and excess $\xi(G) \stackrel{\text{def}}{=\!=} |E(H)|-\frac{3}{2}|V(H)|$.
  4. Degree $\deg_H(v)$ of vertex v in graph H. Nick: default empty option in `\dg{H}` – degree in G.
  5. Δ - upper bound on the maximum degree of G.
  6. $\mathrm{dist}(u, v)$ – distance in G between $u, v \in V$.
- Random 2-SAT formula from G
  1. x_v – boolean variable corresponding to the vertex $v \in V(G)$.
  2. The random formula $\Phi(G)$ picks random literals among $\{x_v, \overline{x_v}\}$ and $\{x_u, \overline{x_u}\}$ for each edge $e = (u, v)$.
  3. Nick: default empty option in `\rform{G}` and macros for $\overline{x_v}$.
  4. $\mathbb{1}_{\text{SAT}}(\Phi(G))$ – event that $\Phi(G)$ is SAT, rarely can use $\{\Phi(G)\ \text{SAT}\}$.
  5. Fixed realization of $\Phi(G)$ – φ.
  6. We define the auxiliary graph $\mathrm{Sol}(\varphi)$ over SAT assignments of φ with $V(\mathrm{Sol}(\varphi)) \subseteq \{0,1\}^V$: two SAT assignments are adjacent in this graph if they differ in exactly one coordinate.
- General conventions
  1. Pr, $\mathbb{E}$ – probability and expectation. Write $\mathbb{E}_\Phi$ to indicate the source of randomness.
- Specific notations from Section 2
  1.
- Specific notations from Section 3
  1.
- Specific notations from Section 4
  1.

Symbol	Meaning	Defined in
R, $B_R(v)$, M_R	a radius; the ball of radius R around v; the bound on its number of vertices	§1.1
ρ	the fraction in "(R, ρ)-locally tree-like"	Def. 1.1
ε	the excess parameter: $\xi(G) \geq \varepsilon n$	Thm. 1.2
δ, (H)	excess density; the standing hypothesis	§1.4
R_1, η	radius of the balls that allow a local contradiction; fraction of vertices with such a ball	Lem. 1.7, 1.8, Cor. 1.9
S, τ	the free variables of a subcube; the values of its fixed variables	§1.5
$X(\varphi)$, $\chi(\cdot)$	the cubical complex of subcubes inside $\mathrm{Sol}(\varphi)$; Euler characteristic	§1.5
F, T, K	the forest of the deletion experiment; number of deferred edges; number of forest components that contain a deferred edge	§1.5
H_p, p	H with each edge kept independently with probability p; the retention probability	§1.6
ℓ, $\mathcal{L}_\ell(H, p)$	path length; probability that a random vertex starts a kept path of length ℓ	§1.6
R_0, ρ_1	the thresholds for R and ρ	Thm. 1.12
ℓ_0, ρ_0, γ	the path length and fraction used for T; the exponential rate	Lem. 1.14

1 Introduction

1.1 The general framework

Nick: not accurate subsection title: it is basically a summary of notations, not the proof framework

An adversary picks a finite multigraph $G = (V, E)$; parallel edges and self-loops are allowed. Write

$$n := |V|, \qquad m := |E|.$$

Each vertex v carries a Boolean variable x_v, and each edge uv carries one of the four clauses

$$x_u \vee x_v, \qquad x_u \vee \neg x_v, \qquad \neg x_u \vee x_v, \qquad \neg x_u \vee \neg x_v.$$

In other words, each end of the edge gets a sign, which decides whether its variable appears negated. The two signs of an edge are independent and uniform, and different edges are independent. A self-loop at v carries a clause on x_v alone, again with two independent uniform signs. The conjunction of all m clauses is a random 2-CNF formula, which we denote by $\Phi(G)$.

Thus the graph is chosen by the adversary, and only the signs are random; this is what *semi-random* means here. A bound on $\Pr(\Phi(G) \text{ SAT})$ must therefore hold for every graph that the adversary can pick, and not only for a typical one.

For a finite graph H, let $c(H)$ be its number of connected components, where each isolated vertex counts as one. Define

$$\nu(H) := |E(H)| - |V(H)| + c(H)$$

to be the total cycle rank of H, that is, the number of independent cycles, and

$$\xi(H) := |E(H)| - \frac{3}{2}|V(H)|$$

to be the excess of H. Positive excess is the same as edge density above 3/2. We write $\deg_H(v)$ for the degree of a vertex v in H; a self-loop at v adds 2 to it.

For $v \in V(G)$ and an integer $R \geq 0$, let $B_R(v)$ be the subgraph induced by all vertices u with graph distance $\mathrm{dist}(u, v) \leq R$. If the maximum degree of G is at most Δ, the layer at distance $j \geq 1$ from v has at most $\Delta(\Delta - 1)^{j-1}$ vertices, so

$$|V(B_R(v))| \leq M_R, \qquad M_R := 1 + \Delta \sum_{j=0}^{R-1} (\Delta - 1)^j.$$

We use M_R throughout as the standard bound on the size of a radius-R ball.

Definition 1.1 (Locally tree-like)**.** Let $R \geq 1$ be an integer and $\rho > 0$. A finite graph G is (R, ρ)-*locally tree-like* if at most $\rho|V(G)|$ vertices v have a radius-R neighborhood $B_R(v)$ that is not a tree.

Finally, for a fixed 2-CNF formula φ, let $\mathrm{Sol}(\varphi)$ be the set of its satisfying assignments. We always regard $\mathrm{Sol}(\varphi)$ as a graph, the *solution graph* of φ, in which two assignments are adjacent when they differ in exactly one variable. Its number of connected components, $c(\mathrm{Sol}(\varphi))$, is the number of solution clusters of φ. In this way $c(\cdot)$ always counts the components of a graph, and $c(\mathrm{Sol}(\varphi)) = 0$ when φ is unsatisfiable.

1.2 Random 2-SAT

Nick: The section starts as "relation to random 2-SAT", then states some of our results and does a warm up lemma (first-moment bound). 3 different things, unclear how are they related

In the classical random 2-SAT model the graph is random as well: a formula on n variables is built from m clauses drawn independently and uniformly. This model has a sharp threshold at clause density 1: the formula is satisfiable with high probability when $m/n < 1$, and unsatisfiable with high probability when $m/n > 1$. The threshold was found by Chvátal–Reed [**?**] and Goerdt [**?**], and the width of the transition window was determined by Bollobás, Borgs, Chayes, Kim and Wilson [**?**].

Since a clause involves two variables, a 2-CNF formula can be viewed as a multigraph with a sign on each edge Nick: sign on an edge? you probably meant "randomly chosen signs of edge variables": the

vertices are the variables, and the edges are the clauses. The ratio m/n is then the edge density of this multigraph. In the classical model the multigraph itself is uniformly random, so the threshold says that a uniformly random multigraph of density above 1 already makes the formula unsatisfiable.

Nick: the next text up to statement of (Main theorem) seems to be out of scope for this subsection. It makes sense to group it with preliminary results on prob. is greater than 0 if density is higher than 3/2. This paper asks what happens when an adversary chooses the multigraph and only the signs are random. We show that the threshold then moves from density 1 to density 3/2. Up to density 3/2, the adversary can choose a graph for which the formula is satisfiable under every sign pattern. Above 3/2, no choice of graph helps: for every graph of density at least $3/2 + \varepsilon$, the formula is unsatisfiable with high probability.

The density 3/2 cannot be lowered. Two variables have four assignments, and a clause on them rules out exactly one of these. Hence three parallel edges between two vertices always leave at least one satisfying assignment, whatever the signs are. This graph has two vertices and three edges, so its density is 3/2, and the same is true for any number of disjoint copies of it. Thus there are graphs with arbitrarily many vertices for which

$$\frac{m}{n} = \frac{3}{2}, \qquad \Pr(\Phi(G) \text{ SAT}) = 1.$$

Our main result says that the adversary cannot do better than this.

Theorem 1.2 (Main theorem)**.** *Fix $\varepsilon > 0$. Then every finite multigraph G with n vertices and*

$$\xi(G) \geq \varepsilon n$$

satisfies

$$\Pr(\Phi(G) \ SAT) \leq e^{-\Omega(n)},$$

where the implied constant depends only on ε. In particular the satisfiability probability tends to 0 as $n \to \infty$.

A first attempt is to count satisfying assignments. This gives a bound of the same form as Theorem 1.2 with a very short proof, but only when the density is at least 2.41.

Lemma 1.3 (First-moment bound)**.** *Every finite multigraph G with n vertices and $m \geq 2.41\, n$ edges satisfies*

$$\Pr(\Phi(G) \ SAT) \leq e^{-\Omega(n)}.$$

Proof. Fix one assignment. For each edge, exactly one of the four sign choices makes its clause false under this assignment, so the clause is true with probability 3/4. Since the signs of different edges are independent, the assignment satisfies $\Phi(G)$ with probability $(3/4)^m$, and linearity of expectation over the 2^n assignments gives $\mathbb{E}\,|\operatorname{Sol}(\Phi(G))| = 2^n(3/4)^m$. If $\Phi(G)$ is satisfiable then $|\operatorname{Sol}(\Phi(G))| \geq 1$, hence

$$\Pr(\Phi(G) \text{ SAT}) \leq \mathbb{E}\,|\operatorname{Sol}(\Phi(G))| = \exp\left[n \log 2 - m \log \tfrac{4}{3}\right].$$

Since

$$2.41 > \frac{\log 2}{\log(4/3)} \approx 2.409,$$

the exponent is at most $-\big(2.41 \log \frac{4}{3} - \log 2\big)n$ whenever $m \geq 2.41\, n$, and $2.41 \log \frac{4}{3} - \log 2 > 0$. □

1.3 Overview of the proof

For densities between 3/2 and 2.41 the first-moment bound fails, and we need a different argument. The proof has three parts. Here we explain the idea of each part and how the parts fit together; the next three subsections give more details.

Part 1: reduction to difficult graphs (Section 2). Some graphs are easy. Suppose that G contains a large number (linear in n) of disjoint subgraphs of constant size, and that each of them has at least one sign pattern that makes its clauses unsatisfiable. Then each subgraph gives a contradiction with constant probability. Since these events are independent, with probability $1 - e^{-\Omega(n)}$ at least one of them happens, and then the formula is unsatisfiable. In Section 2 we show that every graph either has this structure or can be turned into a *difficult graph*, and that this change does not decrease the SAT probability. Specifically, the difficult graph has no loops; its minimum and maximum degree are, respectively, ≥ 2 and

a constant $\leq \Delta$; its excess is linear in n; and the graph is locally tree-like for most of its vertices, i.e., the small neighborhoods (a ball of constant radius) for almost all of its vertices are trees.

Part 2: from satisfiability to a random experiment on the graph (Section 3). For difficult graphs we cannot rely on the local argument, because almost all small neighborhoods are trees, and a 2-CNF whose graph is a forest is satisfiable under every sign pattern. We therefore need a global argument, which makes this section the most interesting part of the proof. The central idea of this section is to count the number of solution clusters instead of the number of solutions. This number is at least 1 whenever the formula is satisfiable, so its expectation is an upper bound on $\Pr(\Phi(G) \text{ SAT})$. The main identity, Theorem 1.10, computes this expectation exactly in terms of the following random edge-deletion experiment on G. We go through the edges of G in a fixed order and build a forest. If the two endpoints of an edge are not yet connected in the forest, we add the edge to the forest with probability 0.5 (and remove the edge with the remaining 0.5 probability). If the pair of vertices are already connected, adding the edge would form a cycle, so we mark this edge and deffer its coin toss. Let T be the number of deferred edges, and let K be the number of connected components of the final forest that contain a deferred edge (the precise definitions are given in Section 1.5). The main identity says that the expected number of solution clusters $\mathbb{E}[c(\mathrm{Sol}(\varphi))]$ equals $\mathbb{E}[2^K(3/4)^T]$. On the left-hand side the expectation is over the random signs of φ, while on the right-hand side it is over the random coins of the experiment with only one coin toss per each edge (instead of two coin tosses per clause in φ). The main identity gives a purely graph-theoretic view of the problem: to bound the satisfiability probability, we only need to study K and T on the graph G.

Part 3: the bounds on K and T (Section 4). For a difficult graph we show that K is small for every outcome of the experiment, while T is large with high probability. A forest component that contains a deferred edge also contains a cycle of G, formed by this edge and the forest path between its endpoints, and different components give disjoint cycles. A locally tree-like graph cannot contain many disjoint cycles, because it has few short ones and there is no room for many long ones, so K is small. For T, suppose that we also flip the coins of the deferred edges. Then every edge is kept with probability $1/2$, and since the edges kept before form a forest, every cycle of the kept subgraph uses a deferred edge. This is where the density $3/2$ comes in. Density above $3/2$ means average degree above 3, so a path that enters a vertex can, on average, leave it along more than 2 other edges. If each edge is kept with probability $1/2$, then on average more than one of these ways survives, so long paths survive and the kept subgraph still has linear cycle rank. This makes T large. As a result, the factor $(3/4)^T$ beats the factor 2^K, and the expectation is exponentially small.

Combining the three parts proves Theorem 1.2.

1.4 The graph-theoretic analysis

In this subsection $\Delta < \infty$ is a degree bound and $\delta > 0$ is an excess density. Both are constants that do not depend on n; their final values are fixed in Lemma 1.4 and depend only on ε.

(H). G has no self-loops, every vertex satisfies $2 \leq \deg_G(v) \leq \Delta$, and $\xi(G) \geq \delta n$.

We call a graph *difficult* if it satisfies (H) and is (R, ρ)-locally tree-like. This subsection deals with all other graphs: for each of them we either prove the conclusion directly or reduce it to a difficult graph. The reduction has four steps.

Step 1: bounded degree. If G has at least $2.41n$ edges, the first-moment bound of Lemma 1.3 already gives the conclusion, so we may assume that G has fewer edges.

Lemma 1.4 (Bounded-degree reduction)**.** *Fix $\varepsilon > 0$. There are constants $\Delta < \infty$ and $\delta > 0$ depending only on ε such that for every graph G with n vertices, fewer than $2.41n$ edges and $\xi(G) \geq \varepsilon n$, one can construct a finite multigraph H with*

$$|V(H)| \geq \frac{\varepsilon}{\Delta} n, \qquad 2 \leq \deg_H(v) \leq \Delta, \qquad \xi(H) \geq 2\delta |V(H)|,$$

and $\Pr(\Phi(G) \text{ SAT}) \leq \Pr(\Phi(H) \text{ SAT})$.

First we may delete vertices of degree 0 or 1, since an isolated variable does not occur in the formula, and a variable that occurs in only one clause can always be set to satisfy it. Both deletions increase the

excess, so they cost nothing. Next, each vertex of large degree is replaced by a path of copies of bounded degree, and the copies are forced to take the same value. Every split adds one vertex and one edge and therefore lowers the excess by 1/2; if the target degree is large enough, the total loss is less than half of the initial excess.

The difficult point is that two clauses are needed to force two neighboring copies to take the same value, while the new graph has only one edge between them. The proof in Section 2.1 solves this by first flipping the variable of each copy at random and then keeping one of the two clauses at random. This produces exactly the right distribution of signs, and removing a clause can only make the formula easier to satisfy.

Step 2: components of nonpositive excess.

Lemma 1.5 (Positive-excess component reduction)**.** *Let G be a finite multigraph with maximum degree at most Δ. Delete every connected component C with $\xi(C) \le 0$, and call the result H. Then every vertex of H has the same degree in H as in G,*

$$\xi(H) \ge \xi(G), \qquad |V(H)| \ge \frac{2}{\Delta}\,\xi(G),$$

every component of H without a self-loop has cycle rank at least 3, every component with a self-loop has cycle rank at least 2, and $\Pr(\Phi(G)\ SAT) \le \Pr(\Phi(H)\ SAT)$.

Since the excess of a graph is the sum of the excesses of its components, deleting components of nonpositive excess can only increase the excess, and the degree bound ensures that the remaining graph is not too small. The main gain, however, is the statement about cycle ranks. For a connected graph C, the condition $\xi(C) > 0$ is the same as $\nu(C) > 1 + |V(C)|/2$. If C has no self-loop, then $|V(C)| \ge 2$, hence $\nu(C) > 2$, and since the cycle rank is an integer, $\nu(C) \ge 3$. If C has a self-loop, then $|V(C)| = 1$ is possible, and we only get $\nu(C) \ge 2$. Steps 3 and 4 need exactly these two bounds.

Step 3: self-loops. The main identity of Section 1.5 holds only for graphs without self-loops, so we must get rid of them. Deleting a self-loop deletes a clause, which can only make the formula easier to satisfy, but it lowers the excess by 1. This is why Steps 1 and 2 keep the excess density 2δ: we can afford to lose δn of the excess. If there are at most δn self-loops, we delete them all, and the excess density goes down from 2δ to δ. This may create vertices of degree 0 or 1 and components of nonpositive excess, so we delete them again as in Steps 1 and 2.

If there are more than δn self-loops, we cannot delete them all, but in this case they help us. A self-loop at v carries one of the clauses x_v, $\neg x_v$ and $x_v \vee \neg x_v$, so with probability 1/2 it fixes the value of x_v, and many such strong local constraints lead to many local contradictions. More precisely, we show that linearly many balls allow a contradiction. For this we need a simple test that tells us when a connected graph has a sign pattern that makes its formula unsatisfiable. The following lemma gives such a test in terms of the cycle rank, and it will be used again in Step 4.

Lemma 1.6 (Unsatisfiable support)**.** *Let G be a finite connected multigraph. Then G supports an unsatisfiable 2-CNF, that is, there is a sign pattern on its edges for which the resulting formula is unsatisfiable, in either of the following cases:*

(1) G has no self-loops and $\nu(G) \ge 3$;

(2) G has a self-loop and $\nu(G) \ge 2$.

For simple graphs the first case is due to Karve–Hirani [**?**, Corollary 19]; parallel edges and self-loops need separate arguments, which we give in Section 2.2. The idea is that, with suitable signs, cycles let a formula carry a forced value around the graph and come back to a contradiction. A self-loop with suitable signs fixes the value of its variable by itself, so one more independent cycle is enough. Note that the lemma only gives sufficient conditions, and without self-loops cycle rank 2 is not always enough: three parallel edges between two vertices have cycle rank 2, but they are satisfiable under every sign pattern.

The next two lemmas, one for Step 3 and one for Step 4, are proved by the same idea. We start from linearly many seeds: the vertices with a self-loop in Lemma 1.7, and disjoint cycles in Lemma 1.8. If several seeds lie close together, a ball around them has enough cycle rank to allow a contradiction. If a seed is alone but its whole component is small, then a ball contains this component, and by Step 2 the component has enough cycle rank. In the remaining case the seed is alone and its component reaches far

away, so the seed has a large neighborhood without other seeds. These neighborhoods are disjoint, so there are only few such seeds. The radius R_1 of the balls is chosen inside each lemma; in Lemma 1.8 it is larger than R.

Lemma 1.7 (Self-loop reduction)**.** *Fix $\Delta < \infty$ and $\delta > 0$. There are $R_1 < \infty$ and $\eta > 0$ depending only on them with the following property. Let G be an n-vertex multigraph with maximum degree at most Δ in which every component containing a self-loop has cycle rank at least 2. If G has more than δn self-loops, then at least ηn vertices v have a ball $B_{R_1}(v)$ that contains a self-loop and has cycle rank at least 2.*

By case (2) of Lemma 1.6, each of these balls allows a contradiction. Here two self-loops close together are already enough, since a self-loop needs only one more independent cycle.

Step 4: local tree structure. After Step 3 the graph has no self-loops, and since Step 2 was applied again, every component has cycle rank at least 3. If the graph is locally tree-like, it is difficult, the reduction is finished, and the graph is treated in the next two subsections. Otherwise the next lemma gives linearly many balls without self-loops and with cycle rank at least 3, and by case (1) of Lemma 1.6 each of them allows a contradiction.

Lemma 1.8 (Local-tree reduction)**.** *Fix $\Delta < \infty$, an integer $R \geq 1$ and $\rho > 0$. There are $R_1 < \infty$ and $\eta > 0$ depending only on them with the following property. Let G be an n-vertex graph without self-loops, with maximum degree at most Δ, in which every component has cycle rank at least 3. If G is not (R, ρ)-locally tree-like, then at least ηn vertices v have a ball $B_{R_1}(v)$ with cycle rank at least 3.*

If the graph is not locally tree-like, then linearly many vertices see a cycle nearby, and a greedy choice among them gives linearly many disjoint cycles, which are the seeds. Here three cycles must lie close together, because two cycles give only cycle rank 2, which is not enough without self-loops. So a seed with at most one other seed nearby is handled by the component argument.

The vertices given by Lemma 1.7 or Lemma 1.8 are exactly what the following statement needs.

Corollary 1.9 (Packing local contradictions)**.** *Fix $\Delta < \infty$, an integer $R_1 \geq 1$ and $\eta > 0$. Suppose G has n vertices and maximum degree at most Δ, and that at least ηn vertices v satisfy one of the following:*

(1) $B_{R_1}(v)$ has no self-loop and $\nu(B_{R_1}(v)) \geq 3$;

(2) $B_{R_1}(v)$ has a self-loop and $\nu(B_{R_1}(v)) \geq 2$.

Then, with the implied constant depending only on Δ, R_1, η,

$$\Pr(\Phi(G) \text{ } SAT) \leq e^{-\Omega(n)}.$$

To prove the corollary, we choose the centers greedily so that the balls are disjoint. Each choice removes only a bounded number of candidates, so we get linearly many disjoint balls. By Lemma 1.6, each of these balls has a sign pattern for which its clauses are unsatisfiable, and since a ball has a bounded number of edges, its random signs equal this pattern with probability bounded below by a positive constant. Disjoint balls have disjoint edges, so these events are independent, and the probability that none of them happens is exponentially small.

Combining the steps. Let G satisfy $\xi(G) \geq \varepsilon n$. Steps 1 and 2 turn it into a graph with degrees in $[2, \Delta]$ and excess density at least 2δ, in which every component has cycle rank at least 3, or at least 2 if it has a self-loop; neither step decreases the satisfiability probability. Step 3 either finishes the proof by packing or removes the self-loops, after which the excess density is at least δ. Step 4 either finishes the proof by packing or gives a difficult graph, with the radius and fraction fixed in Section 1.6. Since every step keeps a constant fraction of the vertices, all the exponential bounds remain exponential in the original n. The details are in Section 2 and Section 4.7.

In the end, therefore, we are left with a difficult graph. For such a graph we cannot rely on the packing argument, because a ball that is a tree allows no contradiction and almost all balls of the fixed radius are trees. The next two subsections deal with this case.

1.5 The expected number of solution clusters

We first explain why the first-moment bound is weak. It counts satisfying assignments, but satisfying assignments come in clusters. Suppose that an assignment satisfies φ, and that every clause containing

some variable is also satisfied by its other literal. Then flipping this variable gives another satisfying assignment, and if several such variables share no clause, they can be flipped in every combination, which gives a whole subcube of solutions. As a result, the number of satisfying assignments can be exponentially larger than the number of really different solutions.

Counting clusters avoids this problem. Since $c(\mathrm{Sol}(\varphi)) \geq 1$ whenever φ is satisfiable, we have

$$\Pr(\Phi(G) \text{ SAT}) \leq \mathbb{E}\, c(\mathrm{Sol}(\Phi(G))),$$

and $c(\mathrm{Sol}(\varphi))$ does not depend on the size of the clusters. The difficulty is that $c(\mathrm{Sol}(\varphi))$ depends on the connectivity of the whole solution graph, so there is no direct way to compute its expectation. The identity below solves this problem.

Since the graph given by the reduction has no self-loops, we only need the identity in this case, and throughout this subsection we assume that G has no self-loops.

Run the following random edge-deletion experiment on G, independently of the signs. Fix an order of the edges, start with the empty graph F, and look at the edges one at a time. If the endpoints of the current edge are not yet connected in F, flip a fair coin right away and add the edge to F if the coin says keep. If the endpoints are already connected, do not flip the coin yet, but defer the edge to the end. In this way F stays a forest during the whole first stage. Write

$$T := \text{number of deferred edges}, \qquad K := \text{number of forest components containing a deferred edge.}$$

Theorem 1.10 (Main identity)**.** *For every finite self-loop-free multigraph G,*

$$\mathbb{E}_\Phi\, c(\mathrm{Sol}(\Phi(G))) = \mathbb{E}\left[2^K \left(\frac{3}{4}\right)^T\right].$$

Corollary 1.11. *For every finite self-loop-free multigraph G,*

$$\Pr(\Phi(G)\ \mathit{SAT}) \leq \mathbb{E}\left[2^K \left(\frac{3}{4}\right)^T\right].$$

Proof. If $\Phi(G)$ is satisfiable, then $c(\mathrm{Sol}(\Phi(G))) \geq 1$. Hence $\Pr(\Phi(G) \text{ SAT}) \leq \mathbb{E}_\Phi\, c(\mathrm{Sol}(\Phi(G)))$, and Theorem 1.10 gives the claim. □

Theorem 1.10 is proved in Section 3. The proof rests on three observations, which we now describe one by one.

The first observation is that the cluster count of a 2-CNF is an Euler characteristic. View the solution set $\mathrm{Sol}(\varphi)$ inside the Boolean cube $\{0,1\}^V$, and record not only the satisfying assignments but also every subcube all of whose assignments satisfy φ. This turns the solution set into a cubical complex $X(\varphi)$. For a general complex, the number of connected components and the Euler characteristic are different numbers, but for a 2-CNF they are equal:

$$c(\mathrm{Sol}(\varphi)) = \chi(X(\varphi)).$$

The reason is a slicing argument. Setting one variable to 0 and to 1 cuts the solution set into two layers, and the assignments that work for both values of this variable connect the two layers. Because every clause has only two literals, two different components of this overlap never lie in the same component of a layer. Hence each component of the overlap joins its own pair of components, one in each layer.

The second observation is that an Euler characteristic is a local alternating sum. A subcube is given by a set $S \subseteq V$ of free variables together with the values $\tau \in \{0,1\}^{V \setminus S}$ of the other, fixed variables, and by definition

$$\chi(X(\varphi)) = \sum_{S \subseteq V} \sum_{\tau} (-1)^{|S|}\, \mathbb{1}\{\text{the subcube given by } S, \tau \text{ lies in } \mathrm{Sol}(\varphi)\},$$

that is, we count points, minus segments, plus squares, and so on. This is the key step: the left-hand side is global, but each indicator on the right is local, and its expectation is easy to compute. A subcube lies in $\mathrm{Sol}(\varphi)$ exactly when every clause is satisfied by all of its assignments. For random signs, the probability that a given clause has this property depends only on which of its two variables are free, not on the

values τ. If we color the free variables blue and the fixed variables red, then an edge is satisfied by the whole subcube with probability 3/4 when both endpoints are red, 1/2 when exactly one is blue, and 0 when both are blue. As a result, $\mathbb{E}\, c(\mathrm{Sol}(\Phi(G)))$ is a signed sum over all red-blue colorings of the vertices. In each term, every red vertex gives a factor 2 (the two values of a fixed variable), every blue vertex gives a factor -1, and every edge gives its factor 3/4, 1/2 or 0. Because of cancellation, this sum can be exponentially small, even though the number of assignments is exponentially large.

The third observation is that the deletion experiment computes this signed sum. Delete each edge independently with probability 1/2, and give a deleted edge the weight 1 and a kept edge the weight 1/2, 0 or -1 according to its color type. The averages are exactly the three numbers 3/4, 1/2 and 0:

$$\tfrac{1}{2}\big(1+\tfrac{1}{2}\big)=\tfrac{3}{4}, \qquad \tfrac{1}{2}(1+0)=\tfrac{1}{2}, \qquad \tfrac{1}{2}(1-1)=0.$$

We insert this deletion into the sum and exchange it with the sum over colorings. Once a deletion outcome is fixed, a kept red-blue edge gives 0, so a coloring contributes only if every component of the kept subgraph has a single color. For one component of the forest, we add the all-red term and the all-blue term and average over the deferred coins; the result is 1 if the component has no deferred edge, and $2(3/4)^t$ if it has $t \geq 1$ deferred edges. Taking the product over all components gives $2^K(3/4)^T$.

Two features of Theorem 1.10 are worth pointing out. First, it is an equality rather than an inequality, so nothing is lost when we pass to the right-hand side. Second, the right-hand side does not involve the formula at all: the question whether $\Phi(G)$ is satisfiable becomes the question whether, in a random experiment on the graph G alone, K is small and T is large. The only loss in this part is the inequality $c(\mathrm{Sol}(\varphi)) \geq 1$ for satisfiable φ.

It remains to prove an upper bound on K and a lower bound on T.

1.6 The analysis for difficult graphs

We now treat difficult graphs, that is, graphs that satisfy (H) and are (R,ρ)-locally tree-like; this is what the reduction above produces. The radius R and the fraction ρ were left open there, and we fix them here.

Definition 1.1 is used here with two different radii, a large radius for the bound on K and a fixed path length for the bound on T; the end of this subsection explains how they fit together.

Theorem 1.12. *There are $R_0 = R_0(\Delta,\delta)$ and $\rho_1 = \rho_1(\Delta,\delta) > 0$ such that the following holds. If $R \geq R_0$, $\rho \leq \rho_1$, and the n-vertex graph G satisfies* (H) *and is (R,ρ)-locally tree-like, then*

$$\mathbb{E}\, c(\mathrm{Sol}(\Phi(G))) = \mathbb{E}\Big[2^K \big(\tfrac{3}{4}\big)^T\Big] \leq 2e^{-\gamma n/2},$$

where $\gamma = \gamma(\Delta,\delta) > 0$ is the rate of Lemma 1.14.

We prove this in Section 4 from the following two bounds, which are of very different types: the first is deterministic and purely local, while the second is probabilistic and comes from the excess.

Lemma 1.13 (Upper bound on K)**.** *If the n-vertex graph G is (R,ρ)-locally tree-like, then every outcome of the deletion experiment satisfies*

$$K \leq \left(\rho + \frac{1}{2R}\right) n.$$

A forest component that contains a deferred edge also contains a cycle of G, namely this edge together with the tree path between its endpoints. Since different components are disjoint, these cycles are vertex-disjoint, and hence K is at most the largest number of vertex-disjoint cycles in G. Long cycles, with more than $2R$ vertices, are few: there are fewer than $n/(2R)$ disjoint ones. A short cycle, of length at most $2R$, lies inside the radius-R ball of each of its vertices, so a locally tree-like graph has at most ρn disjoint ones. Thus K is only a small fraction of n when R is large and ρ is small. Note that this bound uses neither the degree bound nor the excess.

Lemma 1.14 (Exponential moment of T)**.** *There are an integer $\ell_0 = \ell_0(\Delta,\delta)$, $\rho_0 = \rho_0(\Delta,\delta) > 0$ and $\gamma = \gamma(\Delta,\delta) > 0$ such that the following holds. If G satisfies* (H) *and is (ℓ_0,ρ_0)-locally tree-like, then*

$$\mathbb{E}\Big[\big(\tfrac{3}{4}\big)^T\Big] \leq 2e^{-\gamma n}.$$

We state this lemma for $\mathbb{E}[(3/4)^T]$ rather than as a tail bound, because this is the form used in the proof of Theorem 1.12. There the factor 2^K is at most $2^{(\rho+1/(2R))n}$, so we only need $\mathbb{E}[(3/4)^T]$ to decay at a faster exponential rate.

The proof is based on a percolation estimate. For a graph H and $p \in [0,1]$, let H_p be the random subgraph obtained by keeping each edge of H independently with probability p. If we also flip the coins of the deferred edges, then every edge of G is kept independently with probability $1/2$, and the kept edges form $G_{1/2}$. Since the edges kept in the first stage form a forest, every cycle of $G_{1/2}$ uses a deferred edge, and therefore

$$T \;\geq\; \nu(G_{1/2}).$$

It is thus enough to show that $G_{1/2}$ has linearly many independent cycles.

This is where the density $3/2$ appears. Positive excess means $m \geq (\frac{3}{2}+\delta)n$, so the average degree is at least $3+2\delta$. A path that enters a vertex v can leave it in $\deg_G(v)-1$ ways, and we call this number the branching at v; the average branching is therefore at least $2+2\delta$. If each edge is kept with probability p, long paths with branching b survive when $p\,b > 1$ and die out when $p\,b < 1$. The deletion experiment uses $p = 1/2$, so the condition becomes $b > 2$, which means average degree above 3, or density above $3/2$. In other words, the threshold in Theorem 1.2 is exactly the density at which percolation with the fair coin of the deletion experiment becomes supercritical. The example with three parallel edges fits this picture: it has degree 3 and branching exactly 2, which is the critical case.

To get from supercriticality to linearly many cycles, we use one auxiliary quantity, the probability $\mathcal{L}_\ell(H,p)$ that after percolation a uniformly random vertex is the start of a path of length ℓ. Positive excess gives a lower bound on this probability by a constant that does not depend on ℓ. A small cycle rank, on the other hand, gives an upper bound, since a graph with few independent cycles is close to a forest. On a finite forest of bounded degree, percolation with a fixed retention probability $p < 1$ rarely keeps a long path from a random vertex. The two bounds can hold together only if the cycle rank of $G_{1/2}$ is linearly large on average, and a bounded-differences inequality then turns this bound on the expectation into an exponential bound.

Here the local-tree condition is used differently than in the bound on K: it ensures that most vertices have a tree as their neighborhood up to the distance we need. This makes a second-moment computation on paths possible, because two paths that separate inside a tree ball never meet again inside this ball.

The order of the constants. The constants are chosen in the following order, and each of them depends only on the ones before it.

(1) ε is given. Lemma 1.4 then fixes Δ and δ, which depend only on ε.

(2) Lemma 1.14 gives ℓ_0, ρ_0 and γ, which depend only on Δ and δ.

(3) Theorem 1.12 chooses R_0 large and ρ_1 small in terms of ℓ_0, ρ_0 and γ. These are the values of R and ρ used in the reduction.

(4) Finally, Lemmas 1.7 and 1.8, applied with these values, give R_1 and η for Corollary 1.9.

The important point is step (2): the rate γ does not depend on R or ρ. The bound on K loses a factor $2^{(\rho+1/(2R))n}$, so we need $(\rho+\frac{1}{2R})\log 2 < \gamma$. Since γ is fixed before R and ρ, we can take R large and ρ small enough. We also take $R_0 \geq \ell_0$ and $\rho_1 \leq \rho_0$, so that by Lemma 4.1 a graph that is (R_0,ρ_1)-locally tree-like is also (ℓ_0,ρ_0)-locally tree-like, and Lemma 1.14 applies.

1.7 Notation

The following table collects the symbols that are used throughout the paper.

1.8 Organization of the paper

Section 2 contains the graph theory: Section 2.1 proves the bounded-degree reduction; Section 2.2 proves the unsatisfiable-support lemma, the positive-excess component reduction and the packing statement, Corollary 1.9; and Sections 2.3 and 2.4 prove the self-loop reduction and the local-tree reduction. Section 3 proves the main identity, Theorem 1.10. Section 4 treats the graphs that survive all of the reductions: it proves the two bounds, Lemmas 1.13 and 1.14, deduces Theorem 1.12, and combines everything into Theorem 1.2. The two estimates on long paths used there are proved in Appendices A and B.

2 Graph-Theoretic Analysis

This section is pure graph theory. We first remove the vertices of large degree, so that only bounded-degree graphs remain. After that, every case uses the same idea: a small subgraph that allows a contradiction, that is, a small subgraph with at least one sign pattern for which its clauses alone are unsatisfiable. Since such a subgraph has few edges, the random signs hit this pattern with probability bounded below by a positive constant. We use two kinds of small subgraphs, dumbbells and graphs of cycle rank at least 3, and the last two subsections show that a graph which survives the reductions contains linearly many disjoint ones. The random signs then hit the pattern on at least one of them with probability exponentially close to 1.

2.1 Eliminating vertices of large degree

The first reduction turns an arbitrary graph of linear excess into a graph whose degrees lie in a fixed interval, while the dense case is handled by the first-moment bound, Lemma 1.3.

Lemma 1.4 (Bounded-degree reduction). *Fix $\varepsilon > 0$. There are constants $\Delta < \infty$ and $\delta > 0$ depending only on ε such that for every graph G with n vertices, fewer than $2.41n$ edges and $\xi(G) \geq \varepsilon n$, one can construct a finite multigraph H with*

$$|V(H)| \geq \frac{\varepsilon}{\Delta} n, \qquad 2 \leq \deg_H(v) \leq \Delta, \qquad \xi(H) \geq 2\delta |V(H)|,$$

and $\Pr(\Phi(G) \text{ SAT}) \leq \Pr(\Phi(H) \text{ SAT})$.

Lemma 2.1 (Deleting low-degree vertices)**.** *Deleting an isolated vertex does not change satisfiability and increases the excess by $3/2$. Deleting a vertex of degree 1 together with its only incident edge also does not change satisfiability and increases the excess by $1/2$.*

Proof. An isolated variable does not occur in the formula. If x_v occurs in only one clause, then every assignment of the other variables can be extended by choosing the value of x_v that makes this clause true. The two changes in excess are

$$m - \frac{3}{2}(n-1) - \left(m - \frac{3}{2}n\right) = \frac{3}{2}$$

and

$$(m-1) - \frac{3}{2}(n-1) - \left(m - \frac{3}{2}n\right) = \frac{1}{2}.$$

□

Lemma 2.2 (Splitting vertices)**.** *Fix an integer $\Delta \geq 4$. Every finite multigraph G can be changed into a finite multigraph H with maximum degree at most Δ. The graph H has the same number of new vertices as new edges:*

$$|V(H)| - |V(G)| = |E(H)| - |E(G)| \leq \frac{2|E(G)|}{\Delta - 2}.$$

Also, there is a coupling of the random signs for which

$$\{\Phi(G) \text{ SAT}\} \subseteq \{\Phi(H) \text{ SAT}\}.$$

Under this coupling, the two signs on every edge of H are still independent and uniform, and different edges are independent.

Proof. Recall that a self-loop adds 2 to the degree. Write $b := \Delta - 2 \geq 2$ for the number of original edge ends that we allow at one copy. For each vertex v with $\deg_G(v) > b$, let

$$k_v = \left\lceil \frac{\deg_G(v)}{b} \right\rceil,$$

and replace v by k_v copies: split the $\deg_G(v)$ edge ends at v into groups of size at most b, give one group to each copy, and join the copies into a path. (The two ends of a self-loop may go to the same copy or to different copies.) The number of added vertices and the number of added path edges are both

$$N := \sum_v (k_v - 1) \leq \sum_v \frac{\deg_G(v)}{b} = \frac{2|E(G)|}{\Delta - 2}.$$

Each copy has at most b original edge ends and at most two path edges, so the maximum degree of H is at most $b + 2 = \Delta$.

We now construct the coupling. Fix one vertex v that was split, and let $1, \dots, k_v$ be its copies along the path. On each path edge ij we first put the two clauses

$$(\neg x_i \vee x_j) \wedge (x_i \vee \neg x_j),$$

which force $x_i = x_j$. Together these clauses force all copies to take the same value, so the new formula is satisfiable exactly when the original one is. Next, we choose an independent fair bit g_i for each copy i and replace x_i by $x_i \oplus g_i$ in every clause, which does not change satisfiability. Finally, on every path edge we keep only one of its two clauses, chosen independently and uniformly. The result is a subformula of the previous one, so it is satisfiable whenever the original formula is.

It remains to check that the resulting signs have the right distribution. Fix one path of copies and a target sign pattern on all its path edges. If we choose g_i at one end of the path and move along the path, then on each edge the target signs and the bit already chosen determine both the clause that must be kept and the bit g_j at the next copy. Thus every target pattern has exactly two preimages, one for each choice of the first bit, and since all preimages are equally likely, the signs on the path edges are uniform. Different paths use independent bits. Finally, the signs on the original edges were uniform and independent of everything else, and given all bits g_i, the xor with these bits keeps them uniform and independent of the path edges. Hence the result has the same distribution as $\Phi(H)$. □

Proof of Lemma 1.4. Since $m < 2.41\, n$, we have $\varepsilon n \le \xi(G) = m - \frac{3}{2}n < 0.91\, n$, so $\varepsilon < 1$. Fix an integer

$$\Delta > 2 + \frac{5}{\varepsilon},$$

so that $\Delta \ge 8$, and apply Lemma 2.2; call the result H'. Let $N := |V(H')| - n$ be the number of added vertices. Since it is also the number of added edges, the excess goes down by $N/2$, and since

$$N \le \frac{2m}{\Delta - 2} < \frac{2 \cdot 2.41}{5}\, \varepsilon n < \varepsilon n,$$

we get

$$\xi(H') = \xi(G) - \frac{N}{2} > \frac{\varepsilon}{2}\, n, \qquad |V(H')| = n + N < (1 + \varepsilon)n.$$

Now we repeatedly apply Lemma 2.1 to delete vertices of degree 0 or 1, and call the resulting graph H. Since this step can only increase the excess and cannot increase any degree or the number of vertices, we get

$$2 \le \deg_H(v) \le \Delta, \qquad \xi(H) \ge \frac{\varepsilon}{2}\, n > \frac{\varepsilon}{2(1 + \varepsilon)}\, |V(H)|.$$

So we may take

$$\delta := \frac{\varepsilon}{4(1 + \varepsilon)}.$$

The degree bound gives

$$\frac{\varepsilon}{2}\, n \le \xi(H) \le |E(H)| \le \frac{\Delta}{2} |V(H)|,$$

so $|V(H)| \ge (\varepsilon/\Delta)n$. Finally, Lemmas 2.2 and 2.1 give $\Pr(\Phi(G) \text{ SAT}) \le \Pr(\Phi(H) \text{ SAT})$, and both Δ and δ depend only on ε. □

2.2 Unsatisfiable support

We say that a finite multigraph H supports an unsatisfiable 2-CNF if there is at least one sign pattern on the edges of H for which the resulting formula is unsatisfiable. This is a property of the graph alone. Every sign pattern on H has positive probability, so it is the same as saying that $\Phi(H)$ is unsatisfiable with positive probability.

We use two sufficient conditions, which are the two cases below. The first is a *dumbbell*, that is, a self-loop joined by a path to a second cycle. With suitable signs each part has one job: the self-loop fixes its variable, the path carries the forced value, and the second cycle gives the contradiction. Without a self-loop we ask for cycle rank at least 3, because cycle rank 2 is not always enough there: three parallel edges between two vertices have cycle rank 2, but they are satisfiable under every sign pattern.

Lemma 1.6 (Unsatisfiable support). *Let G be a finite connected multigraph. Then G supports an unsatisfiable 2-CNF, that is, there is a sign pattern on its edges for which the resulting formula is unsatisfiable, in either of the following cases:*

(1) G has no self-loops and $\nu(G) \geq 3$;

(2) G has a self-loop and $\nu(G) \geq 2$.

Lemma 2.3 (Dumbbells: multigraphs with a self-loop)**.** *Let H be a finite connected multigraph with a self-loop. If*

$$\nu(H) \geq 2,$$

then H supports an unsatisfiable 2-CNF.

Proof. Choose a self-loop e_0 at a vertex u, and put $(x_u \vee x_u)$ on it, which forces $x_u = 1$. Since $\nu(H) \geq 2$, the graph still contains a cycle after e_0 is deleted, and together with a path from u to this cycle, the self-loop and the cycle form a dumbbell. Walk along the path and then around the cycle. On each edge ab of this walk except the last one, put $(\neg x_a \vee x_b)$, where a is the vertex reached first; these clauses carry the value 1 along the path and around the cycle. On the last edge of the cycle put $(\neg x_a \vee \neg x_b)$, which gives a contradiction. (If the cycle is itself a self-loop at a, put $(\neg x_a \vee \neg x_a)$ on it.) Arbitrary clauses on the unused edges complete the formula. □

Lemma 2.4 (Multigraphs without self-loops)**.** *Let H be a finite connected multigraph without self-loops. If*

$$\nu(H) \geq 3,$$

then H supports an unsatisfiable 2-CNF.

Proof. For each edge $e = uv$ of H, insert a new vertex y_e used only for that edge, and replace e by the path $u - y_e - v$. Since parallel edges get different new vertices, the resulting graph $\widetilde{H}$ is simple, and since each subdivision adds one vertex and one edge,

$$\nu(\widetilde{H}) = \nu(H) \geq 3.$$

By Karve–Hirani [**?**, Corollary 19], $\widetilde{H}$ supports an unsatisfiable formula $\widetilde{\varphi}$. On the path $u - y_e - v$ this formula has two clauses: one contains a literal a_e of x_u and a literal of x_{y_e}, and the other contains a literal b_e of x_v and a literal of x_{y_e}. Since x_{y_e} appears only in these two clauses, we can eliminate it. If x_{y_e} has opposite signs in the two clauses, then

$$\exists x_{y_e} \left[(a_e \vee x_{y_e}) \wedge (b_e \vee \neg x_{y_e})\right] \equiv a_e \vee b_e,$$

and the case with the two signs exchanged is the same. If x_{y_e} has the same sign in both clauses, then one value of x_{y_e} satisfies both clauses, and the pair can be replaced by the always-true formula $\top$. After all new variables are eliminated, $\widetilde{\varphi}$ becomes an unsatisfiable formula that uses each edge of H at most once, and adding arbitrary clauses on the unused edges keeps it unsatisfiable. Hence H supports an unsatisfiable formula. □

The two lemmas above prove Lemma 1.6. We now prove the two statements that use it, namely the positive-excess component reduction and the packing statement. Recall from Section 1.1 that when the maximum degree is at most Δ, every ball of radius R has at most M_R vertices.

Lemma 1.5 (Positive-excess component reduction). *Let G be a finite multigraph with maximum degree at most Δ. Delete every connected component C with $\xi(C) \leq 0$, and call the result H. Then every vertex of H has the same degree in H as in G,*

$$\xi(H) \geq \xi(G), \qquad |V(H)| \geq \frac{2}{\Delta}\,\xi(G),$$

every component of H without a self-loop has cycle rank at least 3, every component with a self-loop has cycle rank at least 2, and $\Pr(\Phi(G)\ SAT) \leq \Pr(\Phi(H)\ SAT)$.

Proof. Since the excess of a graph is the sum of the excesses of its components, deleting components with nonpositive excess gives

$$\xi(H) = \xi(G) - \sum_{C:\,\xi(C)\le 0} \xi(C) \ge \xi(G).$$

Because we delete whole components, the degrees of the remaining vertices do not change, and because we only delete clauses, $\{\Phi(G) \text{ SAT}\} \subseteq \{\Phi(H) \text{ SAT}\}$. The degree bound gives

$$\xi(G) \le \xi(H) \le |E(H)| \le \frac{\Delta}{2}|V(H)|.$$

For every connected component C of H,

$$0 < \xi(C) = \nu(C) - 1 - \frac{|V(C)|}{2}.$$

If C has no self-loops, then $|V(C)| \ge 2$, because a single vertex without edges has negative excess; hence $\nu(C) > 2$, and since $\nu(C)$ is an integer, $\nu(C) \ge 3$. If C has a self-loop, then $|V(C)| \ge 1$, so $\nu(C) > 3/2$ and $\nu(C) \ge 2$. □

Corollary 1.9 (Packing local contradictions). *Fix $\Delta < \infty$, an integer $R_1 \ge 1$ and $\eta > 0$. Suppose G has n vertices and maximum degree at most Δ, and that at least ηn vertices v satisfy one of the following:*

(1) $B_{R_1}(v)$ has no self-loop and $\nu(B_{R_1}(v)) \ge 3$;

(2) $B_{R_1}(v)$ has a self-loop and $\nu(B_{R_1}(v)) \ge 2$.

Then, with the implied constant depending only on Δ, R_1, η,

$$\Pr(\Phi(G) \text{ SAT}) \le e^{-\Omega(n)}.$$

Proof. Call the vertices in the hypothesis candidates, and choose centers greedily among them: each time a center is chosen, remove every candidate at distance at most $2R_1$ from it. At most M_{2R_1} candidates are removed each time, and two chosen centers are at distance more than $2R_1$, so their balls of radius R_1 are vertex-disjoint. Thus we obtain at least

$$\frac{\eta n}{M_{2R_1}}$$

pairwise vertex-disjoint balls of radius R_1.

Each ball is connected, because a shortest path from the center to any vertex of the ball stays inside the ball. Hence, by Lemma 1.6, every ball supports an unsatisfiable 2-CNF, and we fix one such sign pattern for each ball. A ball has at most $\Delta M_{R_1}/2$ edges, and each edge has four sign choices, so the random signs on the ball equal this pattern with probability at least $4^{-\Delta M_{R_1}/2} = 2^{-\Delta M_{R_1}}$. Disjoint balls have disjoint edge sets, so these events are independent, and if one of them happens, the formula is unsatisfiable. Therefore,

$$\Pr(\Phi(G) \text{ SAT}) \le \left(1 - 2^{-\Delta M_{R_1}}\right)^{\eta n / M_{2R_1}} \le \exp\!\left(-2^{-\Delta M_{R_1}} \frac{\eta n}{M_{2R_1}}\right).$$

□

2.3 Self-loop reduction

Lemma 1.7 (Self-loop reduction). *Fix $\Delta < \infty$ and $\delta > 0$. There are $R_1 < \infty$ and $\eta > 0$ depending only on them with the following property. Let G be an n-vertex multigraph with maximum degree at most Δ in which every component containing a self-loop has cycle rank at least 2. If G has more than δn self-loops, then at least ηn vertices v have a ball $B_{R_1}(v)$ that contains a self-loop and has cycle rank at least 2.*

The proof only counts vertices and distances.

Proof of Lemma 1.7. Let V_{loop} be the set of vertices that have at least one self-loop. Since each vertex has at most $\Delta/2$ self-loops,

$$|V_{\text{loop}}| > \frac{2\delta}{\Delta}\, n.$$

Choose an integer R_1 large enough that

$$\frac{1}{\lfloor R_1/2 \rfloor + 1} \le \frac{\delta}{\Delta}.$$

Call $v \in V_{\text{loop}}$ good if at least one of the following holds:

(1) v has at least two self-loops;

(2) the ball $B_{R_1}(v)$ contains the whole component of v;

(3) another vertex of V_{loop} is at distance at most R_1 from v.

For every good vertex v, the ball $B_{R_1}(v)$ contains a self-loop and has cycle rank at least 2: in cases (1) and (3) the ball contains two self-loops, and in case (2) it is the whole component of v, which has cycle rank at least 2 by assumption.

We now bound the number of bad vertices. Two bad vertices are at distance more than R_1, so the balls of radius

$$R' := \left\lfloor \frac{R_1}{2} \right\rfloor$$

around them are pairwise vertex-disjoint. Each such ball has at least $R' + 1$ vertices: since $B_{R_1}(v)$ does not contain the whole component of v, there is a shortest path from v to a vertex at distance more than R_1, and its first $R' + 1$ vertices lie in $B_{R'}(v)$. Hence there are at most $n/(R' + 1) \le (\delta/\Delta)n$ bad vertices, and more than

$$\frac{2\delta}{\Delta}\, n - \frac{\delta}{\Delta}\, n = \frac{\delta}{\Delta}\, n$$

good vertices. This proves the lemma with $\eta := \delta/\Delta$. □

Lemma 2.5 (Removing self-loops)**.** *Suppose that G has n vertices and satisfies*

$$2 \le \deg_G(v) \le \Delta, \qquad \xi(G) \ge 2\delta n,$$

and that every connected component containing a self-loop has cycle rank at least 2. Then either

$$\Pr(\Phi(G) \text{ } SAT) \le e^{-\Omega(n)},$$

with the implied constant depending only on Δ, δ, or one can obtain a graph H without self-loops by deleting only clauses and low-degree vertices, such that

$$|V(H)| \ge \frac{2\delta}{\Delta}\, n, \qquad 2 \le \deg_H(v) \le \Delta, \qquad \xi(H) \ge \delta |V(H)|,$$

every connected component of H has cycle rank at least 3, and

$$\Pr(\Phi(G) \text{ } SAT) \le \Pr(\Phi(H) \text{ } SAT)\,.$$

Proof. First suppose that G has more than δn self-loops. Then Lemma 1.7 gives at least ηn vertices whose ball of radius R_1 contains a self-loop and has cycle rank at least 2, and Corollary 1.9 gives the first conclusion.

Now suppose that G has at most δn self-loops, and delete all of them. Each deleted self-loop lowers the excess by 1, so the excess is still at least δn. Next we repeatedly delete vertices of degree 0 or 1 and call the resulting graph H'; by Lemma 2.1, this step does not lower the excess, so $\xi(H') \ge \delta n$. Finally, we apply Lemma 1.5 to H' and call the result H. Then H has no self-loops, its degrees are the same as in H' and hence lie in $[2, \Delta]$, every component has cycle rank at least 3, and

$$\xi(H) \ge \xi(H') \ge \delta n \ge \delta |V(H)|, \qquad |V(H)| \ge \frac{2}{\Delta}\, \xi(H') \ge \frac{2\delta}{\Delta}\, n.$$

Every operation above deletes clauses or keeps satisfiability unchanged, so $\Pr(\Phi(G) \text{ SAT}) \le \Pr(\Phi(H) \text{ SAT})$. □

2.4 Local-tree reduction

Lemma 1.8 (Local-tree reduction). *Fix $\Delta < \infty$, an integer $R \geq 1$ and $\rho > 0$. There are $R_1 < \infty$ and $\eta > 0$ depending only on them with the following property. Let G be an n-vertex graph without self-loops, with maximum degree at most Δ, in which every component has cycle rank at least 3. If G is not (R, ρ)-locally tree-like, then at least ηn vertices v have a ball $B_{R_1}(v)$ with cycle rank at least 3.*

Proof of Lemma 1.8. Since G is not (R, ρ)-locally tree-like, more than ρn vertices v have a ball $B_R(v)$ that is not a tree. We choose centers greedily among these vertices, removing every candidate at distance at most $2R$ each time a center is chosen. This gives

$$k \geq \frac{\rho}{M_{2R}}\, n =: an$$

centers whose balls of radius R are pairwise vertex-disjoint. None of these balls is a tree, so we can fix one cycle in each of them, and these k cycles are pairwise vertex-disjoint.

Fix an integer s, to be chosen at the end, and build an auxiliary graph on the k centers in which two centers are joined if their distance in G is at most s. We call the connected components of the auxiliary graph *cycle groups*, and we set

$$R_1 := 2s + R.$$

Case 1: at least $k/2$ centers lie in cycle groups of size at least 3. Take such a center w. In the auxiliary graph, two other centers can be reached from w in at most two steps, so these two centers are at distance at most $2s$ from w. Since their cycles lie within distance R of them, the ball $B_{R_1}(w)$ contains three of the vertex-disjoint cycles, and its cycle rank is at least 3. This gives at least $k/2$ good centers.

Case 2: at least $k/2$ centers lie in cycle groups of size 1 or 2. Then there are at least $k/4$ such small groups. We choose one center w from each small group and let

$$R' := \left\lfloor \frac{s}{2} \right\rfloor.$$

Centers from different groups are at distance more than s, so their balls of radius R' are pairwise vertex-disjoint. Suppose that for more than half of the chosen centers w, the ball $B_{R'}(w)$ does not contain the whole component of w. Then each of these balls has at least $R' + 1$ vertices, because a shortest path from w to a vertex outside the ball has its first $R' + 1$ vertices inside the ball, and since these balls are disjoint,

$$n \geq \frac{k}{8}(R' + 1) \geq \frac{an}{8}(R' + 1).$$

We choose s so large that $a(R' + 1) > 8$, which makes this impossible. Hence for at least half of the chosen centers w, that is, for at least $k/8$ centers, the ball $B_{R'}(w)$ contains the whole component of w. Since $R' \leq R_1$, the ball $B_{R_1}(w)$ is this component, and its cycle rank is at least 3 by assumption.

In both cases at least $k/8 \geq (a/8)n$ vertices w satisfy $\nu(B_{R_1}(w)) \geq 3$, which proves the lemma with $\eta := a/8$. The constants s, R_1 and η depend only on Δ, R, ρ. □

3 The Expected Number of Solution Clusters

Theorem 1.10 (Main identity). *For every finite self-loop-free multigraph G,*

$$\mathbb{E}_\Phi\, c(\mathrm{Sol}(\Phi(G))) = \mathbb{E}\left[2^K \left(\frac{3}{4}\right)^T\right].$$

The proof has three steps, one in each subsection. We first show that the number of solution clusters equals an Euler characteristic (Section 3.1), then write its expectation as a signed sum over sets S of free variables (Section 3.2), and finally show that the deletion experiment computes this sum (Section 3.3).

Notation of this section. Besides the symbols of Section 1.7, this section uses two new symbols, the face $Q(S, \tau)$ of the Boolean cube and the edge factor $w_S(e)$.

3.1 Euler characteristic

Let φ be a 2-CNF formula whose variables are indexed by a finite set V; from Section 3.2 on, V is the vertex set of G. The solution set $\mathrm{Sol}(\varphi)$ is a set of vertices of the Boolean cube $\{0,1\}^V$, and to define its Euler characteristic we also record the higher-dimensional faces that lie inside it.

For $S \subseteq V$ and an assignment $\tau : V \setminus S \to \{0,1\}$, let

$$Q(S,\tau) := \{\text{all assignments that agree with } \tau \text{ outside } S\}.$$

The variables in S are free, so $Q(S,\tau)$ has $2^{|S|}$ assignments, and we call it a face of dimension $|S|$. For example, a face of dimension 0 is a point, a face of dimension 1 is a line segment, and a face of dimension 2 is a square.

Let $X(\varphi)$ be the set of all faces $Q(S,\tau)$ with

$$Q(S,\tau) \subseteq \mathrm{Sol}(\varphi).$$

In other words, a face is in $X(\varphi)$ if and only if all its assignments satisfy φ. Since every subface of such a face has the same property, $X(\varphi)$ is a finite cubical complex, and its *Euler characteristic* is

$$\chi(X(\varphi)) := \sum_{Q(S,\tau)\in X(\varphi)} (-1)^{|S|}. \tag{3.1}$$

In words, we count points, minus line segments, plus squares, minus cubes, and so on.

Lemma 3.1 (Euler characteristic of the solution graph)**.** *For every 2-CNF formula φ,*

$$c(\mathrm{Sol}(\varphi)) = \chi(X(\varphi)) = \sum_{S\subseteq V} \sum_{\tau: V\setminus S\to\{0,1\}} (-1)^{|S|} \mathbb{1}_{\{Q(S,\tau)\subseteq \mathrm{Sol}(\varphi)\}}. \tag{3.2}$$

The second equality is just the definition (3.1), written as a sum over all faces. The first equality is proved by induction on the number of variables, and the induction step uses the following slicing formula.

Choose a variable x_u of φ, and let φ_0 and φ_1 be the 2-CNF formulas on the other variables obtained by setting $x_u = 0$ and $x_u = 1$ in φ and simplifying. Thus $\mathrm{Sol}(\varphi_0)$ is the set of assignments y of the other variables such that $(y, x_u = 0)$ satisfies φ, and similarly for $\mathrm{Sol}(\varphi_1)$. An assignment y lies in $\mathrm{Sol}(\varphi_0 \wedge \varphi_1) = \mathrm{Sol}(\varphi_0) \cap \mathrm{Sol}(\varphi_1)$ when it works for both values of x_u, and then it gives an edge of the solution graph of φ between $(y,0)$ and $(y,1)$. As always, we view these three sets as graphs.

Lemma 3.2 (Slicing the solution graph)**.** *For every 2-CNF formula φ and every variable x_u,*

$$c(\mathrm{Sol}(\varphi)) = c(\mathrm{Sol}(\varphi_0)) + c(\mathrm{Sol}(\varphi_1)) - c(\mathrm{Sol}(\varphi_0 \wedge \varphi_1)). \tag{3.3}$$

Proof. Write $I := \mathrm{Sol}(\varphi_0 \wedge \varphi_1)$. If we leave out all edges of the solution graph that flip x_u, what remains is two copies, $\mathrm{Sol}(\varphi_0)$ at $x_u = 0$ and $\mathrm{Sol}(\varphi_1)$ at $x_u = 1$, with $c(\mathrm{Sol}(\varphi_0)) + c(\mathrm{Sol}(\varphi_1))$ components in total. Now we add the edges that flip x_u, which are given by the points of I. We will show that each component of I joins its own pair of components, one of $\mathrm{Sol}(\varphi_0)$ and one of $\mathrm{Sol}(\varphi_1)$, so that each component of I lowers the number of components by exactly 1.

The key point is that two different components of I cannot lie in the same component of $\mathrm{Sol}(\varphi_0)$. Suppose that $y, y' \in I$, and that

$$y = z_0, z_1, \ldots, z_k = y'$$

is a path in $\mathrm{Sol}(\varphi_0)$. For each z_i, define a new assignment z_i' by

$$z_i'(v) := \begin{cases} y(v), & y(v) = y'(v), \\ z_i(v), & y(v) \neq y'(v), \end{cases}$$

for every vertex $v \neq u$. In words, on every variable where y and y' agree we change z_i to their common value, and on every other variable we keep the value of z_i.

We first check that $(z_i', 0)$ satisfies φ, and for this we take any literal. If it is x_u or $\neg x_u$, then $(z_i', 0)$, $(y,0)$ and $(y',0)$ give it the same value. If its variable is one where y and y' agree, then $(z_i', 0)$ gives it the same value as $(y,0)$ and $(y',0)$. Otherwise $(z_i', 0)$ gives it the same value as $(z_i, 0)$, and also the same

value as one of $(y, 0)$ and $(y', 0)$. In every case, if the literal is false under $(z'_i, 0)$, then it is false under at least two of

$$(z_i, 0), \qquad (y, 0), \qquad (y', 0).$$

Now suppose that a clause is false under $(z'_i, 0)$, so that all of its literals are false there. The clause has at most two literals, and each of them is false under two of the three assignments above, so one of the three assignments makes all literals of the clause false. This is a contradiction, because all three assignments satisfy φ.

We next check $(z'_i, 1)$ in the same way, using the three satisfying assignments

$$(z_i, 0), \qquad (y, 1), \qquad (y', 1).$$

For a literal of a variable other than x_u the argument is the same, and for the literal x_u or $\neg x_u$, the last two assignments give it the same value as $(z'_i, 1)$, so it is again false under two of the three. Thus $(z'_i, 1)$ also satisfies φ, and $z'_i \in I$.

The assignments z_i and z_{i+1} differ in one variable, and the change above does not increase the number of variables in which they differ. Hence z'_i and z'_{i+1} are equal or differ in one variable, and after removing repeated points,

$$y = z'_0, z'_1, \ldots, z'_k = y'$$

is a path in I. Therefore one component of $\mathrm{Sol}(\varphi_0)$ meets at most one component of I, and by exchanging 0 and 1, the same holds for $\mathrm{Sol}(\varphi_1)$.

As a result, different components of I join different pairs of components, and each of them lowers the number of components by 1. This proves (3.3). □

Proof of Lemma 3.1. We use induction on the number $|V|$ of variables.

If $|V| = 0$, there is only the empty assignment. If it satisfies φ, then the solution graph has one vertex and $X(\varphi)$ has one face, of dimension 0, so $c(\mathrm{Sol}(\varphi)) = \chi(X(\varphi)) = 1$. If it does not satisfy φ, then the solution graph and $X(\varphi)$ are both empty, and both sides are 0.

Now let $|V| \geq 1$, and assume the result for all 2-CNF formulas with fewer variables. Choose a variable x_u, let φ_0, φ_1 be as above, and divide the faces of $X(\varphi)$ into three types:

- faces with $x_u = 0$; they correspond one-to-one to the faces of $X(\varphi_0)$, with the same dimension;
- faces with $x_u = 1$; they correspond one-to-one to the faces of $X(\varphi_1)$, with the same dimension;
- faces in which x_u is free; they correspond one-to-one to the faces of $X(\varphi_0 \wedge \varphi_1)$, with dimension larger by 1.

Since the dimension in the third type is larger by 1, the sign in the Euler characteristic is reversed there, and therefore

$$\chi(X(\varphi)) = \chi(X(\varphi_0)) + \chi(X(\varphi_1)) - \chi(X(\varphi_0 \wedge \varphi_1)). \tag{3.4}$$

By the induction hypothesis, the right-hand side equals $c(\mathrm{Sol}(\varphi_0)) + c(\mathrm{Sol}(\varphi_1)) - c(\mathrm{Sol}(\varphi_0 \wedge \varphi_1))$, and by Lemma 3.2 this is $c(\mathrm{Sol}(\varphi))$. This completes the induction. □

3.2 Red-blue expansion

Now fix a self-loop-free graph $G = (V, E)$. Taking the expectation of Lemma 3.1 gives

$$\mathbb{E}_\Phi\, c(\mathrm{Sol}(\Phi(G))) = \sum_{S \subseteq V} \sum_{\tau : V \setminus S \to \{0,1\}} (-1)^{|S|} \Pr_\Phi(Q(S, \tau) \subseteq \mathrm{Sol}(\Phi(G)))\,. \tag{3.5}$$

Fix a set S in the outer sum, color the free vertices (those in S) blue and the fixed vertices (those in $V \setminus S$) red, and fix an assignment τ of the red vertices. Whether the random clause on an edge holds on the whole face $Q(S, \tau)$ depends on its signs and on τ, but the probability of this event depends only on the colors of the two endpoints:

endpoints	meaning	probability
red–red	both variables are fixed	3/4
red–blue	one variable is free	1/2
blue–blue	both variables are free	0

For an edge e, let $w_S(e)$ be the number in the last column; thus $w_S(e)$ is 3/4, 1/2 or 0 when e has 0, 1 or 2 endpoints in S.

Lemma 3.3 (Red-blue formula)**.** *For every finite self-loop-free multigraph $G = (V, E)$ with n vertices,*

$$\mathbb{E}_\Phi\, c(\mathrm{Sol}(\Phi(G))) = \sum_{S \subseteq V} 2^{n-|S|}(-1)^{|S|} \prod_{e \in E} w_S(e). \tag{3.6}$$

Proof. Fix S. We first show that all $2^{n-|S|}$ choices of τ in (3.5) give the same probability. Let τ' be another assignment of the red vertices, and in every clause flip the sign of each red variable on which τ and τ' differ. This map is a bijection on sign patterns, and it maps the event $Q(S, \tau) \subseteq \mathrm{Sol}(\Phi(G))$ onto the event $Q(S, \tau') \subseteq \mathrm{Sol}(\Phi(G))$. Since the signs are independent and uniform, the map does not change probabilities, and hence the inner sum in (3.5) equals $2^{n-|S|}(-1)^{|S|}$ times the probability for one fixed τ.

For fixed S and τ the clauses on different edges are independent, so this probability is $\prod_{e \in E} w_S(e)$, and summing over S gives (3.6). □

The Euler characteristic has now become a sum over all red-blue colorings, that is, over all sets S. In the next step we show that each edge factor $w_S(e)$ is the average of a random edge deletion.

3.3 Random edge deletion and deferred revelation

We use the deletion experiment of Section 1.5, with its forest F and its numbers T and K.

Lemma 3.4 (Random edge-deletion formula)**.** *For every finite self-loop-free multigraph $G = (V, E)$ with n vertices,*

$$\mathbb{E}\left[2^K \left(\frac{3}{4}\right)^T\right] = \sum_{S \subseteq V} 2^{n-|S|}(-1)^{|S|} \prod_{e \in E} w_S(e).$$

Lemmas 3.4 and 3.3 together give Theorem 1.10.

Proof. Step 1: one edge. Fix a set S, flip an independent fair coin for each edge, and let $Y_e(S)$ be the following random contribution of the edge e: if e is deleted, then $Y_e(S) = 1$, and if e is kept, then $Y_e(S)$ is 1/2, 0 or -1 when e is red–red, red–blue or blue–blue. Averaging over the coin gives

endpoints	$\mathbb{E}\, Y_e(S)$
red–red	$\frac{1}{2}\left(1 + \frac{1}{2}\right) = \frac{3}{4}$
red–blue	$\frac{1}{2}(1 + 0) = \frac{1}{2}$
blue–blue	$\frac{1}{2}(1 - 1) = 0$

These are the three numbers of the previous table, so $\mathbb{E}\, Y_e(S) = w_S(e)$ for every edge e, and since the coins of different edges are independent,

$$\mathbb{E} \prod_{e \in E} Y_e(S) = \prod_{e \in E} w_S(e).$$

Step 2: exchange the sum and the expectation. We insert this into the right-hand side of the lemma; since both the sum and the expectation are finite,

$$\sum_S 2^{n-|S|}(-1)^{|S|} \prod_{e \in E} w_S(e) = \mathbb{E}\left[\sum_S 2^{n-|S|}(-1)^{|S|} \prod_{e \in E} Y_e(S)\right]. \tag{3.7}$$

Before the exchange we fix S and average over the coins; after it we fix the coins and sum over S. For fixed coins, a deleted edge has factor 1 and can be ignored, while a kept red–blue edge has factor 0. Hence a set S contributes only if every component of the kept graph is all red or all blue.

Step 3: one forest component. Now we reveal the coins in the order of the deletion experiment. After the first stage we know the forest F, but the coins of the deferred edges are not yet revealed, and every

deferred edge has both endpoints in the same component of F. Fix the outcome of the first stage, take one component C of F, and let $t(C)$ be the number of deferred edges in C.

The tree C has $|V(C)|-1$ edges, and they are all kept. If C had vertices of both colors, the tree path between two vertices of different colors would contain a red–blue edge, and the term would be 0. So only two colorings of C contribute: all red and all blue. Let W_C be the number of deferred edges in C that are kept at the end; then $W_C \sim \mathrm{Bin}(t(C), 1/2)$, and C contains $|V(C)|-1+W_C$ kept edges.

If C is all red, each of its vertices gives a factor 2 and each kept edge gives a factor $1/2$, so the term is

$$2^{|V(C)|}\left(\frac{1}{2}\right)^{|V(C)|-1+W_C} = 2^{1-W_C}.$$

If C is all blue, each vertex and each kept edge gives a factor -1, so the term is

$$(-1)^{|V(C)|}(-1)^{|V(C)|-1+W_C} = -(-1)^{W_C}.$$

Now we average over the deferred coins of C. If $t(C)=0$, then $W_C=0$, and the sum of the two terms is $2-1=1$. If $t(C)\geq 1$, then

$$\mathbb{E}\big[2^{1-W_C}-(-1)^{W_C}\big] = 2\left(\frac{1+\frac{1}{2}}{2}\right)^{t(C)} - \left(\frac{1-1}{2}\right)^{t(C)} = 2\left(\frac{3}{4}\right)^{t(C)}.$$

This has a simple meaning edge by edge. In the all-red term, a deferred edge gives 1 if deleted and $1/2$ if kept, so $3/4$ on average, while in the all-blue term it gives 1 or -1, so 0 on average. Thus one deferred edge is enough to make the all-blue term vanish on average.

Step 4: all components. Different components of F have different deferred coins, and a contributing coloring chooses all red or all blue for each component independently. The conditional expectation given the first stage is therefore a product over the components. A component with $t(C)=0$ gives 1, there are exactly K components with $t(C)\geq 1$, and $\sum_C t(C) = T$, so the conditional expectation is

$$\prod_{C:\, t(C)\geq 1} 2\left(\frac{3}{4}\right)^{t(C)} = 2^K\left(\frac{3}{4}\right)^T.$$

Finally, the coins revealed in this order have the same joint law as coins flipped all at the start, so taking the expectation over the first stage gives the lemma. □

4 The Analysis for Difficult Graphs

Recall from Section 1.4 that a graph is difficult if it satisfies (H) and is (R,ρ)-locally tree-like; these are the graphs that survive every reduction of Section 2. Since a tree neighborhood allows no contradiction, we cannot rely on the local argument of Section 2. Instead, in this section we bound the right-hand side of the main identity for such graphs, and then we prove the main theorem.

We need the two bounds of Section 1.6, an upper bound on K and an exponential-moment bound on T, which are of different types. The bound on K holds for every outcome of the deletion experiment and only uses that most neighborhoods are trees, while the bound on T is a probability bound that comes from the positive excess.

The bound on T is the harder one. As explained in Section 1.6, we have $T \geq \nu(G_{1/2})$ (see Section 4.4 for the precise coupling), so it suffices to show that $G_{1/2}$ has linear cycle rank. A direct first-moment count does not give this. Since $\nu(H) = |E(H)| - |V(H)| + c(H)$, we have

$$\mathbb{E}\,\nu(G_{1/2}) = \frac{m}{2} - n + \mathbb{E}\,c(G_{1/2}).$$

When the density m/n is below 2, the term $m/2-n$ is negative, so the numbers of edges and vertices alone give no lower bound. Everything then depends on the number of components, which is hard to control directly. For this reason we take an indirect route: we bound the probability of long kept paths from below and from above (Section 4.3), and compare the two bounds (Section 4.4).

4.1 Standing assumptions and constants

In this section and in Appendices A and B, the constants $\Delta < \infty$ and $\delta > 0$ are fixed, and every constant produced below depends only on them; in particular, none of them depends on R or ρ, which are chosen at the end (see Section 1.6). Recall the standing hypothesis:

(H). G has no self-loops, every vertex satisfies $2 \le \deg_G(v) \le \Delta$, and $\xi(G) \ge \delta n$.

We also use Definition 1.1 with several radii, and the next lemma says that a smaller radius is always allowed.

Lemma 4.1 (Monotonicity)**.** *If G is (R,ρ)-locally tree-like and $1 \le R' \le R$, then G is (R',ρ)-locally tree-like.*

Proof. The ball $B_{R'}(v)$ is an induced subgraph of $B_R(v)$, and it is connected, because a shortest path from v to any of its vertices stays inside it. Since a connected subgraph of a tree is a tree, $B_{R'}(v)$ is a tree whenever $B_R(v)$ is. Hence every vertex with a non-tree ball of radius R' also has a non-tree ball of radius R. □

Notation of this section. Besides the symbols of Section 1.7, this section and Appendices A and B use the following notation: the graph F^+ of all kept edges, the retention probability p_0 and the constant ζ of Lemma 4.2, the error term κ of Lemma 4.3, and the cycle-rank density ω of Lemma 4.4. Symbols that appear only in an appendix are introduced there.

4.2 The upper bound on K

Lemma 1.13 (Upper bound on K). *If the n-vertex graph G is (R,ρ)-locally tree-like, then every outcome of the deletion experiment satisfies*

$$K \le \left(\rho + \frac{1}{2R}\right) n.$$

This lemma needs neither the degree bound nor the excess.

Proof. Each forest component that contains a deferred edge gives a cycle, namely the deferred edge together with the tree path between its endpoints, and cycles from different forest components are vertex-disjoint. We call a cycle short if its length is at most $2R$, and long otherwise.

Every vertex of a short cycle has the whole cycle inside its ball of radius R, so this ball is not a tree. Since the short cycles are vertex-disjoint, Definition 1.1 allows at most ρn of them. A long cycle has more than $2R$ vertices, so, again by disjointness, there are at most $n/(2R)$ long cycles. Adding the two bounds proves the lemma. □

4.3 The long-path functional

The bound on T uses the quantity $\mathcal{L}_\ell(H,p)$ of Section 1.6. Recall that H_p keeps each edge of H independently with probability p, and that

$$\mathcal{L}_\ell(H,p) = \Pr(H_p \text{ contains a path of length } \ell \text{ starting from a uniformly random vertex}). \tag{4.1}$$

The vertex is chosen independently of the edges, and the probability is over both the random vertex and the random edges. A *path of length ℓ* always means a self-avoiding path, that is, $\ell+1$ distinct vertices joined by ℓ edges; walks that repeat a vertex are not counted.

Positive excess makes this quantity large, while a small cycle rank makes it small, and together the two bounds below show that the cycle rank must be linearly large.

Lemma 4.2 (Positive excess gives a lower bound for long paths)**.** *There are constants $p_0 = p_0(\delta) \in (0,1)$, $\zeta = \zeta(\Delta,\delta) > 0$ and $\rho_0 = \rho_0(\Delta,\delta) > 0$ with the following property. Let $\ell \ge 1$ be an integer, and let G satisfy* (H) *and be (ℓ,ρ_0)-locally tree-like. Then*

$$\mathcal{L}_\ell(G, p_0/2) \;\ge\; \zeta. \tag{4.2}$$

Lemma 4.3 (Cycle rank gives an upper bound for long paths)**.** *Let $p < 1$. For every $\kappa > 0$ there is an integer $\ell = \ell(\Delta, p, \kappa)$ such that every finite graph H with maximum degree at most Δ satisfies*

$$\mathcal{L}_\ell(H, p) \;\le\; \kappa + 2M_\ell \frac{\nu(H)}{|V(H)|}. \tag{4.3}$$

Lemma 4.2 is proved in Appendix A, and Lemma 4.3 in Appendix B.

Two points are important here. First, ζ does not depend on ℓ, so we may first fix ζ and then choose ℓ in Lemma 4.3 with $\kappa = \zeta/2$. Second, the lower bound uses the retention probability $p_0/2$, while the upper bound is applied with p_0; as the next subsection shows, the factor $1/2$ comes from the deletion experiment.

4.4 From long paths to cycle rank

Let F be the forest of the first stage of the deletion experiment. Now flip also the coins of the deferred edges, and let F^+ be the graph formed by F and the kept deferred edges. Each edge of G is then kept independently with probability $1/2$, so F^+ has the same law as $G_{1/2}$. Note that F^+ is built from the same coins as F and T, which is what makes the following bound hold for every outcome.

Since both endpoints of every deferred edge lie in the same component of F, adding the kept deferred edges does not join components, and $c(F^+) = c(F)$. Also, since F is a forest on all n vertices, $\nu(F) = 0$, and

$$\nu(F^+) = |E(F^+)| - n + c(F^+) = \nu(F) + \big|E(F^+) \setminus E(F)\big| = \big|E(F^+) \setminus E(F)\big|.$$

Every edge of $E(F^+) \setminus E(F)$ is deferred, so every outcome satisfies

$$T \;\ge\; \nu(F^+). \tag{4.4}$$

Next, we thin once more, keeping each edge of F^+ independently with probability p. Given F^+, the probability that a uniformly random vertex starts a path of length ℓ in the result is $\mathcal{L}_\ell(F^+, p)$. On the other hand, each edge of G survives both steps independently with probability $p/2$, so averaging over F^+ gives

$$\mathbb{E}\big[\mathcal{L}_\ell(F^+, p)\big] = \mathcal{L}_\ell(G, p/2). \tag{4.5}$$

Lemma 4.4 (Positive excess survives half-percolation)**.** *There are constants $\omega = \omega(\Delta, \delta) > 0$, an integer $\ell_0 = \ell_0(\Delta, \delta)$, and $\rho_0 = \rho_0(\Delta, \delta) > 0$ with the following property. If G satisfies* (H) *and is (ℓ_0, ρ_0)-locally tree-like, then*

$$\mathbb{E}\,\nu(G_{1/2}) \;\ge\; \omega\, n.$$

Proof. Let p_0, ζ, ρ_0 be the constants of Lemma 4.2, which depend only on Δ, δ. Apply Lemma 4.3 with $p := p_0$ and $\kappa := \zeta/2$, and let ℓ_0 be the resulting path length, which also depends only on Δ, δ.

Assume G satisfies (H) and is (ℓ_0, ρ_0)-locally tree-like. Then Lemma 4.2 applies with $\ell := \ell_0$, and

$$\zeta \overset{(4.2)}{\le} \mathcal{L}_{\ell_0}(G, p_0/2) \overset{(4.5)}{=} \mathbb{E}\big[\mathcal{L}_{\ell_0}(F^+, p_0)\big] \overset{(4.3)}{\le} \frac{\zeta}{2} + 2M_{\ell_0} \frac{\mathbb{E}\,\nu(F^+)}{n}.$$

In the last step we apply (4.3) to $H = F^+$ for every outcome; this is allowed because F^+ has all n vertices of G and maximum degree at most Δ. Rearranging gives $\mathbb{E}\,\nu(F^+) \ge \omega n$ with

$$\omega := \frac{\zeta}{4M_{\ell_0}},$$

which depends only on Δ, δ. □

4.5 The exponential moment of T

Lemma 1.14 (Exponential moment of T). *There are an integer $\ell_0 = \ell_0(\Delta, \delta)$, $\rho_0 = \rho_0(\Delta, \delta) > 0$ and $\gamma = \gamma(\Delta, \delta) > 0$ such that the following holds. If G satisfies* (H) *and is (ℓ_0, ρ_0)-locally tree-like, then*

$$\mathbb{E}\Big[\big(\tfrac{3}{4}\big)^T\Big] \le 2e^{-\gamma n}.$$

We take ℓ_0 and ρ_0 from Lemma 4.4. To turn the bound on the expectation into a tail bound, we use McDiarmid's bounded-differences inequality in the following form: if f is a function of m independent random variables, and changing one of them changes f by at most 1, then

$$\Pr(f \le \mathbb{E}f - t) \le \exp\left(-\frac{2t^2}{m}\right) \qquad (t > 0).$$

Proof. The graph G has $m \le \Delta n/2$ edges, and $F^+ = G_{1/2}$ is determined by m independent fair coins. Changing one coin changes $\nu(F^+)$ by at most 1. Indeed, suppose the coin adds an edge: if the edge joins two vertices in the same component, the cycle rank goes up by 1, and if it joins two components, the number of edges goes up by 1 and the number of components goes down by 1, so the cycle rank does not change.

Lemma 4.4 gives $\mathbb{E}\,\nu(F^+) \ge \omega n$. We apply the inequality above with $f := \nu(F^+)$ and $t := \omega n/2$, and use $m \le \Delta n/2$:

$$\Pr\big(\nu(F^+) < \tfrac{\omega}{2}n\big) \le \exp\left(-\frac{2(\omega n/2)^2}{m}\right) \le \exp\left(-\frac{\omega^2}{\Delta}n\right).$$

By (4.4), the same bound holds for $\Pr\big(T < \frac{\omega}{2}n\big)$.

Now we split according to whether T is small:

$$\mathbb{E}\Big[\big(\tfrac{3}{4}\big)^T\Big] \le \big(\tfrac{3}{4}\big)^{\omega n/2} + \Pr\big(T < \tfrac{\omega}{2}n\big) \le \exp\left(-\frac{\omega}{2}\log\frac{4}{3}\cdot n\right) + \exp\left(-\frac{\omega^2}{\Delta}n\right) \le 2e^{-\gamma n},$$

where

$$\gamma := \min\left\{\frac{\omega}{2}\log\frac{4}{3},\ \frac{\omega^2}{\Delta}\right\} > 0$$

depends only on Δ, δ. □

4.6 Proof of Theorem 1.12

Theorem 1.12. *There are $R_0 = R_0(\Delta,\delta)$ and $\rho_1 = \rho_1(\Delta,\delta) > 0$ such that the following holds. If $R \ge R_0$, $\rho \le \rho_1$, and the n-vertex graph G satisfies* (H) *and is (R,ρ)-locally tree-like, then*

$$\mathbb{E}\,c(\mathrm{Sol}(\Phi(G))) = \mathbb{E}\Big[2^K\,\big(\tfrac{3}{4}\big)^T\Big] \le 2e^{-\gamma n/2},$$

where $\gamma = \gamma(\Delta,\delta) > 0$ is the rate of Lemma 1.14.

Proof. The equality is the main identity, Theorem 1.10, which applies because (H) forbids self-loops.

Let ℓ_0, ρ_0, γ be the constants of Lemma 1.14; they depend only on Δ, δ. Set

$$R_0 := \max\left\{\ell_0,\ \left\lceil\frac{2\log 2}{\gamma}\right\rceil\right\}, \qquad \rho_1 := \min\left\{\rho_0,\ \frac{\gamma}{4\log 2}\right\}.$$

If $R \ge R_0$ and $\rho \le \rho_1$, then

$$\Big(\rho + \frac{1}{2R}\Big)\log 2 \le \frac{\gamma}{4} + \frac{\gamma}{4} = \frac{\gamma}{2}.$$

Moreover, since $R \ge \ell_0$ and $\rho \le \rho_0$, Lemma 4.1 shows that G is (ℓ_0,ρ_0)-locally tree-like, so Lemma 1.14 applies. Since Lemma 1.13 gives $K \le (\rho + \frac{1}{2R})n$ for every outcome, we get

$$\mathbb{E}\Big[2^K\,\big(\tfrac{3}{4}\big)^T\Big] \le 2^{(\rho+\frac{1}{2R})n}\,\mathbb{E}\Big[\big(\tfrac{3}{4}\big)^T\Big] \le 2\cdot e^{\left((\rho+\frac{1}{2R})\log 2-\gamma\right)n} \le 2e^{-\gamma n/2}.$$

□

4.7 Proof of the main theorem

All parts are now available, namely the reductions of Section 2, the identity of Section 3, and Theorem 1.12.

Theorem 1.2 (Main theorem). *Fix $\varepsilon > 0$. Then every finite multigraph G with n vertices and*

$$\xi(G) \geq \varepsilon n$$

satisfies

$$\Pr(\Phi(G) \text{ SAT}) \leq e^{-\Omega(n)},$$

where the implied constant depends only on ε. In particular the satisfiability probability tends to 0 as $n \to \infty$.

Proof of Theorem 1.2. Let Δ and δ be the constants of Lemma 1.4, which depend only on ε, and let R_0 and ρ_1 be the constants of Theorem 1.12 for these Δ, δ. In this proof, H_0, H_1, H_2 are three graphs, and every $\Omega(\cdot)$ hides a constant that depends only on ε.

Step 1. If $m \geq 2.41\, n$, then Lemma 1.3 gives $\Pr(\Phi(G) \text{ SAT}) \leq e^{-\Omega(n)}$ and we are done. Otherwise Lemma 1.4 gives a graph H_0 with

$$|V(H_0)| \geq \frac{\varepsilon}{\Delta} n, \qquad 2 \leq \deg_{H_0}(v) \leq \Delta, \qquad \xi(H_0) \geq 2\delta |V(H_0)|,$$

and $\Pr(\Phi(G) \text{ SAT}) \leq \Pr(\Phi(H_0) \text{ SAT})$.

Step 2. We apply Lemma 1.5 to H_0 and call the result H_1. Its degrees are the same as in H_0, and

$$\xi(H_1) \geq \xi(H_0) \geq 2\delta |V(H_0)| \geq 2\delta |V(H_1)|, \qquad |V(H_1)| \geq \frac{2}{\Delta} \xi(H_0) \geq \frac{4\delta}{\Delta} |V(H_0)| = \Omega(n).$$

Every component of H_1 without a self-loop has cycle rank at least 3, and every component with a self-loop has cycle rank at least 2.

Step 3. We apply Lemma 2.5 to H_1. Either $\Pr(\Phi(H_1) \text{ SAT}) \leq e^{-\Omega(|V(H_1)|)}$, or we get a graph H_2 without self-loops with

$$|V(H_2)| \geq \frac{2\delta}{\Delta} |V(H_1)| = \Omega(n), \qquad 2 \leq \deg_{H_2}(v) \leq \Delta, \qquad \xi(H_2) \geq \delta |V(H_2)|,$$

in which every component has cycle rank at least 3, and $\Pr(\Phi(H_1) \text{ SAT}) \leq \Pr(\Phi(H_2) \text{ SAT})$.

Step 4. If H_2 is not (R_0, ρ_1)-locally tree-like, then Lemma 1.8, with $R = R_0$ and $\rho = \rho_1$, gives a radius R_1 and a fraction η, and Corollary 1.9 gives $\Pr(\Phi(H_2) \text{ SAT}) \leq e^{-\Omega(|V(H_2)|)}$. Otherwise H_2 satisfies (H) and is (R_0, ρ_1)-locally tree-like, so Corollary 1.11 and Theorem 1.12, with $R = R_0$ and $\rho = \rho_1$, give

$$\Pr(\Phi(H_2) \text{ SAT}) \leq \mathbb{E}\, c(\text{Sol}(\Phi(H_2))) \leq e^{-\Omega(|V(H_2)|)}.$$

In every case the bound is exponential in $|V(H_1)|$ or $|V(H_2)|$, both of which are $\Omega(n)$, and all implied constants depend only on ε. Together with the chain $\Pr(\Phi(G) \text{ SAT}) \leq \Pr(\Phi(H_0) \text{ SAT}) \leq \Pr(\Phi(H_1) \text{ SAT}) \leq \Pr(\Phi(H_2) \text{ SAT})$, this gives

$$\Pr(\Phi(G) \text{ SAT}) \leq e^{-\Omega(n)}.$$

□

A Proof of the Positive-Excess Lower Bound

This appendix proves Lemma 4.2. The proof has two independent halves. The first half is a second-moment computation, which shows that if a constant fraction of the starting edges carry a well spread family of paths, then a long path survives the random deletion with constant probability. The second half builds this family from the positive excess.

Notation of this appendix. Besides the symbols of Sections 1.7 and 4, this appendix uses the following notation: the set $\mathcal{P}_\ell(e)$ of candidate paths, the weights w_e, a path prefix π, the branching rate λ, the fraction ψ, and the drift μ and upper bound U of the log-branching walk.

A.1 Flows on the path tree

Fix a directed edge $e = (u, v)$, and let $\mathcal{P}_\ell(e)$ be the set of paths of length ℓ that start at v and whose first step does not go to u. Give each $P \in \mathcal{P}_\ell(e)$ a weight $w_e(P) \geq 0$, and for a path prefix π of length k, write

$$w_e(\pi) := \sum_{P \supseteq \pi} w_e(P)$$

for the total weight of the candidate paths that begin with π.

Counting the candidate paths is not enough. If there are many of them and each edge is kept with probability p, the expected number of kept paths may be large, but the paths may share a long first segment, and then deleting one common edge destroys all of them. We therefore need a family that is *spread out*, and we express this with a flow.

Definition A.1 (Flow)**.** Let $\lambda > 1$ and $\psi \in (0, 1]$. Let $e = (u, v)$ be a directed edge such that $B_\ell(v)$ is a tree. A weight function w_e on $\mathcal{P}_\ell(e)$ is a (λ, ψ)-*flow* if

$$\sum_{P \in \mathcal{P}_\ell(e)} w_e(P) = 1, \qquad w_e(\pi) \leq \frac{\lambda^{-k}}{\psi} \quad \text{for every prefix } \pi \text{ of length } k.$$

If such a w_e exists, we say that e *carries a* (λ, ψ)-*flow.*

Thus a flow is a unit flow in which every prefix of length k has capacity λ^{-k}/ψ. On infinite trees, flows of this kind are used to define the branching number (see Lyons [**?**] and Lyons–Peres [**?**]), but we do not need that theory, because the flows we use are built directly in Lemma A.3. The next lemma is the standard second-moment argument which shows that percolation survives when $p\lambda > 1$.

Lemma A.2 (Deletion calculation for weighted paths)**.** *Suppose that G has n vertices, no self-loops, and $2 \leq \deg_G(v) \leq \Delta$. Let $\lambda > 1$ and $\psi \in (0, 1]$, and assume that at least a ψ fraction of the directed edges carry a (λ, ψ)-flow. If $p\lambda > 1$, then*

$$\mathcal{L}_\ell(G, p) \geq \frac{2\psi^2}{\Delta} \left(1 - (p\lambda)^{-1}\right). \tag{A.1}$$

This lemma does not use the excess, which is needed only in Lemma A.3 to produce the flows.

Proof. First fix a directed edge $e = (u, v)$ that carries a (λ, ψ)-flow, together with such a flow w_e. Keep each edge with probability p, and define

$$Z_e := \sum_{P \in \mathcal{P}_\ell(e)} p^{-\ell} w_e(P) \mathbb{1}_{\{\text{all edges of } P \text{ are kept}\}}.$$

Each candidate path has ℓ edges, so $\mathbb{E}Z_e = \sum_P w_e(P) = 1$.

Take two candidate paths P, Q, and let $k(P, Q)$ be the length of their common prefix. All candidate paths lie in the tree $B_\ell(v)$, where two paths from v that separate never meet again, so P and Q have

exactly $k(P,Q)$ common edges, and both are kept with probability $p^{2\ell-k(P,Q)}$. Therefore

$$\begin{aligned}\mathbb{E}Z_e^2 &= \sum_{P,Q} w_e(P)w_e(Q)p^{-k(P,Q)} \\ &\le \sum_{k=0}^{\ell} p^{-k} \sum_{|\pi|=k} w_e(\pi)^2 \\ &\le \sum_{k=0}^{\ell} p^{-k} \left(\max_{|\pi|=k} w_e(\pi)\right) \sum_{|\pi|=k} w_e(\pi) \\ &\le \frac{1}{\psi}\sum_{k=0}^{\ell}(p\lambda)^{-k} \le \frac{1}{\psi\big(1-(p\lambda)^{-1}\big)}.\end{aligned}$$

For the second line, use $p^{-k(P,Q)} \le \sum_{k=0}^{k(P,Q)} p^{-k}$, and note that P and Q have a common prefix of length k exactly when $k \le k(P,Q)$; then group the pairs (P,Q) by this common prefix π. For the last line, note that every candidate path has exactly one prefix of each length k, so $\sum_{|\pi|=k} w_e(\pi) = 1$. By the Cauchy–Schwarz inequality,

$$\Pr(Z_e > 0) \ge \frac{(\mathbb{E}Z_e)^2}{\mathbb{E}Z_e^2} \ge \psi\left(1-(p\lambda)^{-1}\right),$$

and if $Z_e > 0$, then some candidate path from v has all its edges kept.

Now we average over the directed edges. For $e = (u,v)$, the event $Z_e > 0$ is contained in the event that a kept path of length ℓ starts at v, and a vertex v is the head of exactly $\deg_G(v) \le \Delta$ directed edges, so

$$\sum_{e \text{ carrying a flow}} \Pr(Z_e > 0) \le \Delta \sum_{v\in V} \Pr(\text{a kept path of length } \ell \text{ starts at } v) = \Delta\, n\, \mathcal{L}_\ell(G,p).$$

Since the minimum degree is at least 2, the graph G has at least n edges, that is, at least $2n$ directed edges, and at least $2\psi n$ of them carry a flow. Hence

$$\mathcal{L}_\ell(G,p) \ge \frac{1}{\Delta\, n} \cdot 2\psi n \cdot \psi\left(1-(p\lambda)^{-1}\right) = \frac{2\psi^2}{\Delta}\left(1-(p\lambda)^{-1}\right).$$

□

It remains to produce the flows, and this is where the positive excess is used.

Lemma A.3 (Construction of the flows)**.** *There are constants $\lambda = \lambda(\delta) \in \big(2,\, 2(1+\delta)\big)$, $\psi = \psi(\Delta,\delta) > 0$ and $\rho_0 = \rho_0(\Delta,\delta) > 0$ with the following property. Let $\ell \ge 1$ be an integer, and let G satisfy* (H) *and be (ℓ,ρ_0)-locally tree-like. Then at least a ψ fraction of the directed edges carry a (λ,ψ)-flow.*

Proof of Lemma 4.2. Take λ, ψ, ρ_0 from Lemma A.3. Since $\lambda > 2$, we have $2/\lambda < 1$, so we can fix

$$p_0 := \frac{1}{2}\Big(\frac{2}{\lambda}+1\Big) \in \Big(\frac{2}{\lambda}, 1\Big).$$

Then $(p_0/2)\lambda > 1$, and since Lemma A.3 gives the hypothesis of Lemma A.2, we can apply that lemma with $p = p_0/2$:

$$\mathcal{L}_\ell(G, p_0/2) \ge \frac{2\psi^2}{\Delta}\left(1-\left(\tfrac{p_0}{2}\lambda\right)^{-1}\right) =: \zeta > 0.$$

All of $\lambda, \psi, \rho_0, p_0, \zeta$ depend only on Δ, δ, and none of them depends on ℓ, n or G. This proves Lemma 4.2. □

Here we need $2/\lambda < 1$, that is, $\lambda > 2$, and we can choose such a λ below $2(1+\delta)$ exactly because $\delta > 0$; see (A.5) below.

A.2 Constructing the flows

Proof of Lemma A.3. The walk. We use an auxiliary random walk only to define the weights; the graph G is fixed, and there is no edge deletion here. We write $\Pr_{\mathrm{nb}}$ for the law of the following nonbacktracking walk.

Choose a directed edge $\vec{e}_0 = (v_{-1}, v_0)$ uniformly from all directed edges of G. After reaching $\vec{e}_i = (v_{i-1}, v_i)$, choose $\vec{e}_{i+1}$ uniformly among the $\deg_G(v_i) - 1$ directed edges that leave v_i and are not the reversal of $\vec{e}_i$, and write $\vec{e}_{i+1} = (v_i, v_{i+1})$. Since every vertex has degree at least 2, the walk never gets stuck.

Note that the walk forbids the reversal of the edge just used, not every edge back to v_{i-1}. The two rules differ only when v_{i-1} and v_i are joined by parallel edges, and with the second rule such a step would have fewer than $\deg_G(v_i) - 1$ choices, so the stationarity below would fail. We also need that G has no self-loops, since a self-loop at v_i would give a step with $v_{i+1} = v_i$.

For a fixed directed edge e, we write $\Pr_{\mathrm{nb}}(\cdot \mid e)$ for the law of the walk started with $\vec{e}_0 = e$.

Stationarity. Fix a directed edge (a, b). The directed edges that can move to (a, b) are the $\deg_G(a) - 1$ edges that enter a, except (b, a), and each of them moves to (a, b) with probability $1/(\deg_G(a) - 1)$. Hence if $\vec{e}_i$ is uniform over all directed edges, then

$$\Pr_{\mathrm{nb}}(\vec{e}_{i+1} = (a, b)) = (\deg_G(a) - 1) \cdot \frac{1}{2|E(G)|} \cdot \frac{1}{\deg_G(a) - 1} = \frac{1}{2|E(G)|},$$

and $\vec{e}_{i+1}$ is uniform as well. Since $\vec{e}_0$ is uniform, it follows that every $\vec{e}_i$ is uniform, and since the same rule is used at every step, for all $t \geq 0$ and $j \geq 1$ the blocks

$$(\vec{e}_0, \dots, \vec{e}_{j-1}) \quad \text{and} \quad (\vec{e}_t, \dots, \vec{e}_{t+j-1})$$

have the same joint law. We call this property *stationarity*; note that it does not mean that the steps are independent.

The weights. Let $\lambda > 2$ be the constant fixed in (A.5) below. Call a walk of length ℓ *good* if

$$\prod_{i=0}^{k-1} (\deg_G(v_i) - 1) \geq \lambda^k \qquad \text{for every } 1 \leq k \leq \ell. \tag{A.2}$$

In words, every prefix of a good walk has enough branching. Let $\mathcal{G}$ be the event that the walk is good, let $g(e) := \Pr_{\mathrm{nb}}(\mathcal{G} \mid e)$, and if $g(e) > 0$, define

$$w_e(P) := \frac{\Pr_{\mathrm{nb}}(P \mid e)}{g(e)} \mathbb{1}_{\{P \in \mathcal{G}\}}.$$

Thus w_e is the law of the walk, restricted to good walks and normalized.

Now let $e = (v_{-1}, v_0)$ be a directed edge such that $B_\ell(v_0)$ is a tree. Then the walks of length ℓ started with e are exactly the paths in $\mathcal{P}_\ell(e)$. Indeed, in a tree there is only one edge between v_{-1} and v_0, so forbidding this edge is the same as forbidding the vertex v_{-1}, and in a tree a nonbacktracking walk never comes back to a vertex. Hence w_e is a weight function on $\mathcal{P}_\ell(e)$ with total mass 1. To check the capacity bound, let π be a prefix of length k. If π is not the beginning of a good walk, then $w_e(\pi) = 0$; otherwise (A.2) holds for π, and

$$w_e(\pi) \leq \frac{\Pr_{\mathrm{nb}}(\pi \mid e)}{g(e)} = \frac{1}{g(e)} \prod_{i=0}^{k-1} \frac{1}{\deg_G(v_i) - 1} \leq \frac{\lambda^{-k}}{g(e)}. \tag{A.3}$$

Therefore, if $g(e) \geq \psi$, then e carries a (λ, ψ)-flow, and it remains to find λ and ψ such that at least a ψ fraction of the directed edges $e = (u, v)$ have $g(e) \geq \psi$ and a tree ball $B_\ell(v)$.

Choice of λ. Let $d := 2|E(G)|/n$ be the average degree, so that $d \geq 3 + 2\delta$ by (H). Since $\vec{e}_i$ is uniform, the vertex v_i is chosen with probability proportional to its degree:

$$\Pr_{\mathrm{nb}}(v_i = v) = \frac{\deg_G(v)}{2|E(G)|}.$$

The function $f(s) = s \log(s - 1)$ is convex for $s \geq 2$, since $f''(s) = (s - 2)/(s - 1)^2 \geq 0$. By Jensen's inequality,

$$\mathbb{E}_{\mathrm{nb}} \log(\deg_G(v_i) - 1) = \frac{\sum_v f(\deg_G(v))}{nd} \geq \frac{f(d)}{d} = \log(d - 1) \geq \log\big(2(1 + \delta)\big). \tag{A.4}$$

Fix

$$\lambda = \lambda(\delta) := 2(1+\delta)^{1/4} \in \big(2,\ 2(1+\delta)\big). \tag{A.5}$$

This interval is nonempty because $\delta > 0$, and the bound $d \geq 3 + 2\delta$ in (A.4) and this choice are the only places where the positive excess is used. The upper end $2(1+\delta)$ makes the drift below positive, while the lower end 2 leaves room for the retention probability $p_0/2$.

The drift. Define

$$Y_i := \log(\deg_G(v_i) - 1) - \log\lambda,$$

so that $\sum_{i<k} Y_i \geq 0$ is the same as the inequality in (A.2). Since Y_i is a function of $\vec{e}_i$, the sequence $(Y_i)_{i\geq 0}$ is stationary, and by (A.4),

$$\mathbb{E}_{\mathrm{nb}} Y_i \ \geq\ \log\big(2(1+\delta)\big) - \log\lambda \ =:\ \mu \ >\ 0, \qquad Y_i \leq \log(\Delta - 1) =: U.$$

We need more than a positive average, because every prefix sum up to ℓ must be nonnegative. The following lemma, proved in Section A.3, gives this.

Lemma A.4 (Finite positive-prefix lemma)**.** *Let $(Y_i)_{i\geq 0}$ be a stationary sequence of real random variables, that is, for all integers $t \geq 0$ and $j \geq 1$ the blocks $(Y_0, \dots, Y_{j-1})$ and $(Y_t, \dots, Y_{t+j-1})$ have the same joint law. Assume that*

$$Y_i \leq U, \qquad \mathbb{E}Y_i \geq \mu > 0.$$

Then there is a constant $\psi \in (0, 1/4]$, depending only on U and μ, such that for every integer $\ell \geq 1$,

$$\Pr\left(\sum_{i=0}^{k-1} Y_i \geq 0 \text{ for every } 1 \leq k \leq \ell\right) \geq 4\psi. \tag{A.6}$$

Many good starting edges. By Lemma A.4, the walk is good with probability at least 4ψ, and since the starting edge is uniform,

$$\frac{1}{2|E(G)|} \sum_{e} g(e) \ \geq\ 4\psi,$$

where the sum is over all directed edges. Since $0 \leq g(e) \leq 1$, at least a 2ψ fraction of the directed edges satisfy

$$g(e) \geq 2\psi, \tag{A.7}$$

since otherwise the average would be less than $2\psi + 2\psi = 4\psi$.

The tree condition. Choose $\rho_0 > 0$ with $\Delta\rho_0/2 \leq \psi$. Since G is (ℓ, ρ_0)-locally tree-like, at most $\rho_0 n$ vertices have a non-tree ball of radius ℓ, and at most $\Delta\rho_0 n$ directed edges end at these vertices. There are at least $2n$ directed edges in total, so at most a ψ fraction of them end at such a vertex. Hence at least a ψ fraction of the directed edges $e = (u, v)$ satisfy (A.7) and have a tree ball $B_\ell(v)$, and by (A.3) each of them carries a (λ, ψ)-flow. Since the constants λ, ψ and ρ_0 depend only on Δ, δ, this proves Lemma A.3. □

A.3 Proof of the finite positive-prefix lemma

Proof of Lemma A.4. The two assumptions give $\mu \leq \mathbb{E}Y_0 \leq U$.

A deterministic fact. Let $y_0, \dots, y_{N-1} \leq U$ be real numbers with $W := \sum_{i=0}^{N-1} y_i > 0$, and extend the sequence periodically by $y_{i+N} = y_i$. Set $s_k := \sum_{i=0}^{k-1} y_i$, so that $s_0 = 0$ and $s_{k+N} = s_k + W$, and call a position $i \in \{0, \dots, N-1\}$ *good* if

$$s_{i+k} - s_i = \sum_{j=i}^{i+k-1} y_j \geq 0 \qquad \text{for every } 1 \leq k \leq N.$$

We show that at least W/U positions are good. Let

$$\underline{s}_i := \min_{j \geq i} s_j$$

be the smallest partial sum from position i on; the minimum exists because the partial sums grow by $W > 0$ in every period. Since $\underline{s}_i = \min\{s_i, \underline{s}_{i+1}\}$, we have $\underline{s}_{i+1} \geq \underline{s}_i$. If $\underline{s}_{i+1} > \underline{s}_i$, then $\underline{s}_i = s_i$, hence

$\underline{s}_{i+1} \le s_{i+1} = s_i + y_i \le \underline{s}_i + U$. Also $s_{i+k} \ge \underline{s}_{i+1} > s_i$ for every $k \ge 1$, so position i is good. Thus every step $\underline{s}_{i+1} - \underline{s}_i$ is either 0 or at most U, and it is positive only at a good position. Finally,

$$\sum_{i=0}^{N-1} (\underline{s}_{i+1} - \underline{s}_i) = \underline{s}_N - \underline{s}_0 = W,$$

because $\underline{s}_N = \min_{j \ge 0}(s_j + W) = \underline{s}_0 + W$. Hence at least W/U of these steps are positive, and at least W/U positions are good.

A block with positive sum. Fix ℓ, let $N := \lceil 4U/\mu \rceil \, \ell$, and set $W_N := \sum_{i=0}^{N-1} Y_i$. By stationarity,

$$\mathbb{E}W_N \ge \mu N, \qquad W_N \le UN.$$

Therefore,

$$\Pr\Big(W_N \ge \frac{\mu}{2} N\Big) \ge \frac{\mu}{2U}. \tag{A.8}$$

Indeed, if the probability on the left is a, then $\mu N \le \mathbb{E}W_N \le aUN + (1-a)\mu N/2$, which gives $a \ge \mu/(2U - \mu) \ge \mu/(2U)$.

Many good positions. On the event in (A.8), the deterministic fact with $y_i = Y_i$ gives at least $\mu N/(2U)$ good positions. A good position i with $i + \ell \le N$ gives $\sum_{j=i}^{i+k-1} Y_j \ge 0$ for all $1 \le k \le \ell$, using only terms of the block and no periodic copies, and at most ℓ positions have $i + \ell > N$. Since $N \ge 4U\ell/\mu$, we have $\ell \le \mu N/(4U)$, so at least

$$\frac{\mu N}{2U} - \ell \ge \frac{\mu N}{4U}$$

good positions have $i + \ell \le N$.

Back to position 0. Let J be uniform on $\{0, \dots, N-1\}$ and independent of the sequence, and let $\mathcal{E}_i$ be the event that $\sum_{j=i}^{i+k-1} Y_j \ge 0$ for every $1 \le k \le \ell$. On the event in (A.8), at least a $\mu/(4U)$ fraction of the positions i satisfy $\mathcal{E}_i$, so

$$\Pr(\mathcal{E}_J) \ge \frac{\mu}{4U} \Pr\Big(W_N \ge \frac{\mu}{2} N\Big) \ge \frac{\mu^2}{8U^2}.$$

By stationarity, $\Pr(\mathcal{E}_i) = \Pr(\mathcal{E}_0)$ for every i, and therefore $\Pr(\mathcal{E}_J) = \Pr(\mathcal{E}_0)$. If we set

$$\psi := \frac{\mu^2}{32U^2},$$

then $\Pr(\mathcal{E}_0) \ge 4\psi$, and $\mu \le U$ gives $\psi \le 1/32 \le 1/4$. The constant ψ depends only on U and μ, and not on ℓ. □

B Proof of the Cycle-Rank Upper Bound

We restate the result to be proved.

Lemma 4.3 (Cycle rank gives an upper bound for long paths). Let $p < 1$. For every $\kappa > 0$ there is an integer $\ell = \ell(\Delta, p, \kappa)$ such that every finite graph H with maximum degree at most Δ satisfies

$$\mathcal{L}_\ell(H, p) \le \kappa + 2M_\ell \frac{\nu(H)}{|V(H)|}.$$

The proof first handles forests and then reduces a general graph to a forest by removing $\nu(H)$ edges. Besides the symbols of Sections 1.7 and 4, this appendix uses two new symbols, the forest extinction bound $h(\ell)$ and the separator set Sep.

B.1 From the forest case to a general graph

Lemma B.1 (Uniform extinction in finite forests)**.** *Let $p < 1$. There is a function $h(\ell)$ of the path length, depending only on Δ and p, with*

$$h(\ell) \longrightarrow 0 \qquad (\ell \to \infty),$$

such that every finite forest H with maximum degree at most Δ satisfies

$$\mathcal{L}_\ell(H, p) \le h(\ell). \tag{B.1}$$

Proof of Lemma 4.3. Choose a spanning tree in every connected component of H. Their union is a spanning forest $H' \subseteq H$, and we call the other edges of H extra edges. By the definition of cycle rank, there are exactly

$$|E(H) \setminus E(H')| = \nu(H)$$

extra edges.

Choose a vertex uniformly at random and keep each edge of H with probability p. If the kept graph contains a path of length ℓ from the chosen vertex, then one of two cases holds:

(1) the path uses only edges of H';

(2) the path uses at least one extra edge.

In case (1), the path lies in H'_p. Since the forest H' has the same vertices as H, the chosen vertex is also uniform in H', so by Lemma B.1, case (1) has probability at most $h(\ell)$.

In case (2), we only count which vertices can be the start and ignore whether the edges are kept. If a path of length ℓ from a vertex v uses an extra edge e, then v is at distance at most ℓ from one endpoint of e. So v lies in one of the two balls of radius ℓ around the endpoints of e, which contain at most $2M_\ell$ vertices. Since there are $\nu(H)$ extra edges, at most $2M_\ell\nu(H)$ vertices are possible starts. The start is uniform among the $|V(H)|$ vertices, so case (2) has probability at most

$$2M_\ell \frac{\nu(H)}{|V(H)|}.$$

Adding the two cases gives

$$\mathcal{L}_\ell(H,p) \le h(\ell) + 2M_\ell \frac{\nu(H)}{|V(H)|}. \tag{B.2}$$

Given $\kappa > 0$, choose ℓ so large that $h(\ell) \le \kappa$; this is possible because $h(\ell) \to 0$, and ℓ depends only on Δ, p, κ. Then (B.2) gives (4.3). □

Only the property $h(\ell) \to 0$ is used here, and the exact formula for $h(\ell)$ does not matter. It remains to prove the forest estimate.

B.2 Uniform extinction in finite forests

Proof of Lemma B.1. Cutting the forest. Fix an integer $N \ge 2$. Root each tree of H at any vertex, and go through the vertices from the leaves toward the root, so that every vertex comes after all vertices below it. At each vertex, look at the vertices of its subtree, itself included, that are not yet assigned. If there are at least N of them, put the vertex into a set Sep and assign all of them to it. Since each vertex of Sep gets at least N vertices and different vertices of Sep get disjoint parts, we obtain

$$|\mathrm{Sep}| \le \frac{|V(H)|}{N}. \tag{B.3}$$

After this process, every connected component of $H - \mathrm{Sep}$ has fewer than N vertices.

Small trees. Define

$$g(N) := \sup_{(C,u)} \sum_{v \in V(C)} p^{\mathrm{dist}_C(u,v)},$$

where the supremum is over all trees C with fewer than N vertices and maximum degree at most Δ, and all vertices u of C. Since at most $\min\{N, \Delta(\Delta-1)^{t-1}\}$ vertices are at distance $t \ge 1$ from u, we have

$$g(N) \le 1 + \sum_{t \ge 1} \min\{N, \Delta(\Delta-1)^{t-1}\} p^t, \qquad \frac{g(N)}{N} \longrightarrow 0 \quad (N \to \infty). \tag{B.4}$$

To see the limit, note that if $\Delta = 2$, the sum is bounded by a constant. If $\Delta \ge 3$, we split the sum at $t = \lceil \log_{\Delta-1} N \rceil$; then the whole sum is at most a constant times $\log N + N^{1+\log_{\Delta-1} p}$, and $1 + \log_{\Delta-1} p < 1$ because $p < 1$.

Long paths reach Sep. Suppose H_p contains a path of length N from a vertex v. This path has $N+1$ vertices, so it does not fit into one component of $H - \mathrm{Sep}$, and hence v is joined to Sep by kept edges. A vertex $v \notin \mathrm{Sep}$ can reach Sep only through an edge sa with $s \in \mathrm{Sep}$ and $a \notin \mathrm{Sep}$, where v lies in the

component of a in $H - \mathrm{Sep}$. Since paths in a forest are unique, this happens with probability at most $p^{\mathrm{dist}(a,v)}$. There are at most $\Delta|\mathrm{Sep}|$ such edges sa, and for each of them the sum over v is at most $g(N)$, so after adding the vertices of Sep themselves we get

$$\sum_{v \in V(H)} \Pr(v \text{ is joined to Sep in } H_p) \le |\mathrm{Sep}| + \Delta|\mathrm{Sep}|g(N).$$

Dividing by $|V(H)|$, using (B.3), and taking $N = \ell$ for $\ell \ge 2$, we obtain (B.1) with

$$h(\ell) := \frac{1 + \Delta\, g(\ell)}{\ell} \quad (\ell \ge 2), \qquad h(1) := 1,$$

and $h(\ell) \to 0$ by (B.4); for $\ell = 1$ the bound is trivial. □